European Volatility Under the Microscope

A Machine Learning Approach to EURO STOXX 50 Forecasting

Tobias Persson† & Anton Johanson‡

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Abstract

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Keywords: Volatility Forecasting, Realized Volatility, HAR Model, Realized Quarticity, Variance Risk Premium, High-Frequency Data

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Supervisor: Tobias Sichert, Department of Finance, Stockholm School of Economics
Examinator: Gualtiero Azzalini, Department of Finance, Stockholm School of Economics
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†tobias.ct.persson@outlook.com
‡anton13johanson@gmail.com

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1 Introduction

The question of European financial autonomy, once confined to policy seminars, has in recent years become a central preoccupation of investors, regulators, and policymakers alike. In his landmark 2024 report on *The Future of European Competitiveness* European Commission (2025), Mario Draghi diagnosed the fragmented state of European capital markets as nothing short of an existential challenge, warning that without deep structural reform, including the formation of a Capital Markets Union aimed at creating a single integrated capital market across the European Union, Europe would fail to sustain its productivity, security, and long-term global competitiveness.

Against this backdrop, European equity markets have entered a new phase of elevated relevance to global investors. A structural refuge of capital outflows from US equities toward European capital markets accelerated in 2025, as rising trade policy uncertainty and historic depreciation of the US dollar prompted investors to diversify their geographical holdings. According to Morningstar (2026), European large-cap funds netted €41.2 billion in flows in 2025, a stark increase from the €12.0 billion of outflows in 2024, while US large-cap funds closed with merely €6.2 billion on aggregate. In the longer run, UBS Flood (2025) projects that a base case reallocation of capital amounting to €1.2 trillion into European equities could materialize in the upcoming five years as investors divert there investments from US equities. For all types of investors, both domestic and foreign, the ability to model, measure, and forecast the trajectory of European equity market risk has become an integral element of capital allocation, risk management, and long-term portfolio performance.

Volatility forecasting is the art of capturing and measuring the level of uncertainty prevalent across financial markets. More accurate and precise volatility forecasts translate directly into better calibrated risk models, resulting in more informed decision-making for capital investments. The origins of the volatility forecasting literature trace back to the original works of Engle (1982) and Bollerslev (1986), who introduced the initial ARCH and GARCH models respectively to forecast volatility movements in financial markets. With time, access to high-frequency trading data and stronger compute allowed practitioners to expand this strand of literature, by introducing new specifications like the Heterogeneous Autoregressive (HAR) set of models originally derived by Corsi (2009). While the standard HAR models and its extensions have become salient benchmarks in volatility forecasting, advanced computers and programs have introduced advanced machine learning algorithms that can capture nonlinearities and complex patterns between correlated variables not previously observed by the conventional set of linear models in the forecasting literature.

Ongoing talks on forming a European Capital Markets Union, together with rising foreign capital infows to European markets, warrants a close examination of the volatility dynamics that prevail in Europe equity markets, along with the models best suited to forecast future movements. The main objective of our paper is therefore to extend the analysis of Christensen et al. (2023) to the European context, by testing the out-of-sample volatility forecasting performance for a set of different models on the EURO STOXX 50 index. Importantly, our extension is motivated by three key considerations.

First of all, the empirical literature on ML-based volatility forecasting remains heavily concentrated on US equity markets. To the best of our knowledge, no study has undertaken a comprehensive comparison between HAR-type models and modern machine learning approaches for the EURO STOXX 50 index. This gap in the literature is striking given the central role of the index as the primary benchmark for European equity markets performance, broadly comparable to the S&P500 in the US. Moreover, the EURO STOXX 50 index serves as the underlying asset for the VSTOXX index, Europe's principal measure of implied volatility, analogous to the VIX index in the US context.

Second of all, the macro-financial structure of the Eurozone is materially distinct from that of the US, with important implications for volatility dynamics and their predictability. Unlike the US, which constitutes a relatively integrated single-economy system, the EURO STOXX 50 index mirrors a multi-sovereign environment characterized by heterogenous fiscal conditions, sovereign risk differentials, and varying degrees of financial fragmentation, which have been shown to influence macroeconomic stability and asset price behavior (Lane (2012), Corsetti et al. (2013)). As a result, realized volatility in European equity markets is shaped not only by firm-level and aggregate factors, but also by supranational monetary policy conducted by the European Central Bank (ECB) Bohl et al. (2008), systematic financial stress as captured by the Composite Indicator of Systemic Stress (CISS) index (Hollo et al. (2012)), and heightened exposure to external shocks, including energy price fluctuations (Katsampoxakis et al. (2022)). Moreover, the open nature of the Eurozone economy implies that exchange rate movements and global spillovers from Asian and US trading sessions can play a more pronounced role in European equity markets relative to the US context (Wang et al. (2020), Golosnoy et al. (2015)). Taken together, these structural differences suggest that volatility forecasting models developed for US markets may not directly generalize to the European setting without explicitly accounting for these additional macro-financial variables highly relevant in the European context.

Finally, the single-index setting of our paper differs structurally from the cross-sectional approach of Christensen et al. (2023), whereby we study one target index series as opposed to the twenty nine individual stocks studied in Christensen et al. (2023). Moreover, contrary to Christensen et al. (2023), our cleaned sample spans a longer period from January 1999 to August 2020, thereby including the Dot-com Bubble (2000 - 2002), the Global Financial Crisis (GFC, 2008 - 2009), the European Sovereign Debt Crisis (Euro Crisis, 2010 - 2012), a prolonged low-volatility expansion period (2016 - 2019), and the Covid-19 market turmoil of March 2020. These five macroeconomic regimes provides a demanding and informative testing ground for volatility forecasting model evaluation.

The paper is outlined as follows. Section (2) provides a historical and chronological overview of how the literature on volatility forecasting has evolved over time. Section (3) puts forth the data gathering and cleaning process while also motivating the selection of variables that we have choosen to include in our feature sets. Section (4) offers detailed insight into the functionalities of each and every forecasting model, as well as an explanation for the methodological approach we use to assess out-of-sample performance. Section (5) presents the empirical results gathered from the different feature sets we use in our approach. Finally, sections (6) and (7) conclude our paper by discussing our findings, their implications, potential limitations of our paper, and explores new avenues for future research on the topic of volatility forecasting.

2 Literature Review

This section reviews literature on volatility forecasting by presenting a chronological overview of model development. We begin by introducing the autoregressive models, upon which modern research is developed, and transition towards modern approaches based on high-frequency data, including linear HAR models and novel machine learning techniques. The section concludes by examining the existing empirical evidence for European equity markets and outlining how this study contributes to the broader volatility forecasting literature.

2.1 From ARCH and GARCH to High-frequency Data

The origins of the literature of volatility forecasting can be traced back to the contribution of Engle (1982), who introduced a new class of autoregressive conditional heteroskedastic (ARCH) models, which allows the conditional variance to vary over time as a function of past squared residuals while the unconditional variance remains constant. Bollerslev (1986) generalized this class of autoregressive processes to the GARCH model to allow for a longer and more flexible lag structure. As an infinite order ARCH model, the GARCH framework models conditional variance as a function of both past squared residuals and conditional variances, and have been found to outperform high-order ARCH specifications, thus constituting an important benchmark in volatility modeling.

Estimated on daily returns, however, the forecast accuracy of the standard GARCH and its various extensions (see Lawrence R. Glosten (1993), Nelson (1991), Schwert (1990), Robert F. Engle (1993), Zakoian (1994), Sentana (1995), Hentschel (1995), Baillie (1996)) is inherently limited as these parametric models disregard the dynamics observed within intraday price information when treating the entire trading day as a single observation. Recognizing these limitations, Andersen (2001) proposed exploiting the information embedded in high-frequency intraday data by introducing realized volatility as a model-free and consistent estimator of the latent variation in daily returns, which can be directly observed and not just modeled. While realized volatility is highly skewed, Andersen (2001) found that its logarithmic transformation is approximately Gaussian and highly persistent, thus making it well-suited for standard time series modeling. Building on this conceptual shift, Andersen (2003) demonstrated that simple linear models, applied to realized volatility outperforms traditional GARCH and stochastic volatility models, highlighting the key advantages of high-frequency data in volatility forecasting.

2.2 The Linear HAR Universe

Following the contributions of Andersen (2001) and Andersen (2003) in establishing realized volatility as a persistent and accurate predictor of latent volatility, Corsi (2009) introduced the heterogeneous autoregressive (HAR) model as a parsimonious and intuitive framework for volatility forecasting. Grounded in the Heterogeneous Market Hypothesis (HMH) presented by Ulrich A. Müller (1997), which asserts that financial markets consists of heterogeneous traders rather than a homogeneous group investors, the HAR model approximates the long-memory characteristics of realized volatility through an additive cascade of volatility components measured over daily, weekly, and monthly horizons, and it has been shown to provide more robust forecasts compared to other autoregressive models with more complicated lag structures.

The HAR framework has since been extended in various directions to account for more salient features observed in financial data. In addition to the baseline HAR model, Corsi (2009) also take the logarithm of realized volatility (LogHAR) to avoid negative volatilities and increase the accuracy of forecasts as the distribution converge to normal. Corsi and Renò (2012) extend the standard HAR model by decomposing realized volatility into a continuous variation component and a jump component (LevHAR), and show that the forecasting performance can be improved by including a persistent leverage effect. Patton and Sheppard (2015) account for the fact that negative returns have a larger impact on future volatility compared to positive returns by decomposing the total variation in the standard HAR framework into its positive and negative return components, realized semivariance (SHAR). The authors find that negative semivariance contains significantly more predictive power than its positive counterpart, indicating that distinguishing between these two can lead to higher forecast accuracy. Bollerslev et al. (2016a) explicitly account for the variance of the measurement error in RV estimators, instead of assuming it is constant over time, to construct more efficient forecasts (HARQ) that exhibits higher persistence during tranquil periods and quicker mean-reversion during periods of distress.

2.3 Economic Drivers of Volatility

Related work has focused on the economic drivers of stock market volatility and whether the forecast accuracy of HAR-type models can be improved by incorporating additional sources of information. In particular, Paye (2012) documents that the persistence in realized volatility can be more accurately modeled by conditioning on macroeconomic and financial variables, suggesting that volatility dynamics is not only determined by its lagged values but also by broader economic fundamentals. This is in line with the findings in Charlotte Christiansen (2012), Robert F. Engle and Sohn (2013), and Nonejad (2017), where macroeconomic indicators played a significant role in predicting future volatility. Extending this argument beyond traditional macroeconomic variables, Yudong Wang (2018) show that crude oil volatility is a strong predictor of stock-market volatility in the short-term and demonstrate that augmenting on this new variable improves forecast accuracy over the autoregressive benchmark.

Taufiq Choudhry (2016) consider the opposite direction when investigating the relationship between stock market volatility and real economic activity in the US, Canada, Japan, and the UK, and further extend the analysis by examining potential spillover effects. Based on a combination of linear and non-linear causality tests, the findings indicate that stock market volatility is an important short-term predictor of the business cycle in all four countries, and that this relationship is more pronounced during periods of heightened volatility. Furthermore, the study identifies significant spillover effects, indicating that stock market volatility and economic activity in the US may also predict economic fundamentals in the other three markets. Collectively, this line of research suggests that the autoregressive specifications of the standard HAR model may omit information that is relevant when forecasting future volatility and conditioning on a broader set of macroeconomic fundamentals can improve forecast accuracy.

2.4 Machine Learning: Regularization and Nonlinearity

The application of machine learning (ML) in volatility forecasting is motivated by the challenges faced by traditional econometric techniques, such as linear regressions, when dealing with large and high-dimensional datasets, variable selection, and potential nonlinear relationships as model complexity increases. As emphasized by Varian (2014), the growing availability of more powerful data manipulation tools that take these problems into account can be used to model more complex relationships than those captured by simple linear models.

Several studies have focused on comparing the predictive performance of ML and the linear HAR framework. Audrino and Knaus (2016) examine whether the least absolute shrinkage and selection operator (LASSO) can be used as a model selection tool to recover the lag structure of the HAR model and to what extent it can be used for forecasting realized volatility. They find no distinguishable differences in forecast accuracy across the two models. Furthermore, their findings suggest that LASSO can recover the lag structure of the HAR only under the assumption that HAR is the true model but fails to do so on real data, indicating that the true volatility dynamics may be more complex than those captured by the strict lag structure of the HAR framework. Moreover, the out-of-sample forecasting performance of LASSO and HAR is comparable. Yi Ding (2021) further extend the analysis of forecasting performance of the LASSO relative to the standard HAR model and find that the LASSO approach significantly outperforms the HAR model, which is attributed to the flexible lag structure of the LASSO.

Other studies have also examined the effectiveness of regularization techniques. Yu Wei (2020) explore the use of ridge regression to forecast the realized volatility of gold future prices in China, where HAR models augmented by implied volatility indices are used to predict the volatility of the Chinese gold futures market. The ridge regression is documented to outperform standard HAR models by aiding potential overfitting and multicollinearity problems among the predictors. Using tree-based models, Chuong Luong (2018) apply the random forests (RF) algorithm to high frequency returns data from the S&P ASX 200 with the purpose of forecasting the direction and magnitude of realized volatility, and find that the RF approach was able to improve forecast accuracy. In related work, Rangan Gupta (2022) show that the accuracy of forecasting the price of crude oil increase at intermediate and long horizons when random forests are used as the primary estimation technique relative to the AR-type benchmark. Similarly, Elie Bouri (2021) document an improvement in forecast accuracy across random forest configurations compared to the HAR benchmark when predicting the realized volatility of Bitcoin returns following the US-China trade war.

Boosting techniques, which generates decision trees sequentially rather than independently to allow for continuous learning, are applied to a large set of financial and macroeconomic predictors in Stefan Mittnik (2015) to identify the main drivers of stock market volatility. The componentwise gradient boosting approach is shown to generate superior out-of-sample volatility forecasts both in the short- and longer-run compared to the GARCH benchmark, thus indicating that non-linear elements drive market risk. Marcelo Fernandes (2014) examine the time-series properties of the CBOE VIX and evaluate traditional HAR model against parametric and semiparametric HAR processes augmented by additional macroeconomic variables, where the semiparametric model is approximated through neural networks. Following the highly persistence nature of the VIX index, the standard HAR model is found to performs in line with the extended specifications with macroeconomic variables, while the neural network show no evidence of superior out-of-sample forecasting at different forecasting horizons.

A comprehensive comparison of these approaches is provided by Christensen et al. (2023), which serves as the primary benchmark for this thesis. The authors evaluate the out-of-sample performance for a wide range of machine learning algorithms, including inear regularization methods, tree-based models, and neural networks, against the benchmark HAR family of models in the context of volatility forecasting. The study is conducted in a cross-sectional setting using high-frequency data for 29 constituents of the DOJ Jones Industrial Average (DJIA) index over the period from January 2001 to December 2017, rather than focusing on a single aggregate index. In addition to the standard daily, weekly, and monthly realized volatility components, the information set is expanded to include firm-level characteristics and macroeconomic variables. This two-set framework allows the authors to assess the extent to which machine learning methods improve forecasting accuracy relative to HAR-type models and to identify the key predictors driving these improvements.

Based on an out-of-sample MSE evaluation for one-day ahead realized volatility forecasts, Christensen et al. (2023) document clear differences in performance across model classes. First, regularization methods, such as LASSO, ridge regression, and elastic net, provide consistent improvements over the HAR benchmark, particularly when a large set of predictors is included, thus effectively controlling for overfitting and allowing for variable selection. Second, while tree-based models, including bagging, random forests, and gradient boosting, underperform relative to the HAR lineage for the lagged variance set, the predictive accuracy is increased for the unrestricted dataset, reflecting their ability to capture non-linearities and interactions among predictors. Finally, the neural network models exhibit further gains in forecast accuracy across both feature settings, although the magnitude of these improvements varies with the model specification and additional hidden layers to the network does not necessarily result in accelerated performance. Overall, these findings indicate that while machine learning models tend to outperform traditional HAR specifications, particularly when a large set of predictors is available, traditional models remain competitive in more parsimonious settings.

2.5 Prior Research on European Volatility Forecasting

While a substantial body of literature has examined volatility forecasting using both traditional econometric models and more recent machine learning approaches, the majority of prior research is centered on domestic equity indices or individual constituents rather than broader, cross-country indices (Elias Søvik Gunnarsson (2024); Juan Mansilla-Lopez (2025); Radmir Mishelevich Leushuis (2026)). Although Yi Ding (2021) include the UK FTSE 100 and German DAX 30 as benchmarks when comparing the performance of the LASSO model relative to the HAR framework and Patton and Sheppard (2015) incorporate nternational indices, such as the FTSE 100 and EURO STOXX 50 alongside S&P 500 data, in the analysis of the leverage effect and the impact of signed returns, the empirical evidence on remains relatively fragmented. For instance, Barry Harrison (2012) apply AR and MA models to forecast daily volatility for stock market exchanges in Central and and Eastern Europe, with the aim of analyzing return behavior in emerging markets as opposed to developed Western European markets. In a different approach, Fabrizio Cipollini (2019) decompose realized volatility for five major European equity indices into common and market-specific components and examine their contribution to EURO STOXX 50 volatility when incorporated into a HAR framework. Finally, Javier Parra-Domínguez (2024) evaluate the effectiveness of various machine learning techniques, including CNN, ANN, and LSTM models, in predicting the closing price of the EURO STOXX 50. However, their focus on daily price prediction differs from our study, which instead considers the forecasting of realized volatility.

3 Data Gathering and Cleaning

The empirical analysis in this paper builds on the application of 1-minute high-frequency intraday trading data for the EURO STOXX 50 index, obtained from the Stockholm School of Economics. The initial sample, before the data cleaning process, spans the period from January 1998 to August 2020.

The EURO STOXX 50 index is composed of the 50 largest and most liquid companies in the Eurozone, covering about 20 different industry sectors. Therefore, the EURO STOXX 50 is often regarded as the European continent's equivalent to that of the S&P500 index, making it a natural benchmark for evaluating the state and performance of European equity markets.

Before moving forward with volatility forecasting, we perform a careful and extensive data cleaning process on the initial data set. We proceed to remove weekends and any trading day that has more than 50% of missing trading data. In our initial sample, there are only 164 days with more than 50% of missing price data, spread across the entire sample period. After careful inspection, we do not consider there to be a major survivorship bias concern with regard to this data cleaning step, given that the 164 trading days removed do not appear to be systematically concentrated in any particular period of our full sample. To account for minor data gaps, we forward-fill and backward-fill the remaining price gaps within each day if there are any. Note that the missing price data in some intraday windows appears rather as idiosyncratic and random data quality issues as opposed to larger, concentrated, data gaps. As such, this approach preserves days with only small interruptions while excluding days whose price record is too incomplete to support reliable realized variance estimation. To understand how we specifically set the range for the intraday trading window, see Subsection (3.1).

Once we merge all variables together in our extensive feature sets, see Subsection (3.3), we acknowledge that the VSTOXX index only starts recording data from 1999-01-04 onwards. Consequently, our final data set is constrained by this limitation. Therefore, we end up with a fully cleaned sample spanning the entire period from the 1999-01-05 to 2020-08-06.

Note that, because we examine the continuous EURO STOXX 50 index series, contrary to a fixed basket of constituents, our data inherits index rebalancing by construction. This differs from Christensen et al. (2023), who work with individual firms. The trade-off is that our index series reflects the evolving European equity landscape but may absorb jumps at rebalancing dates.

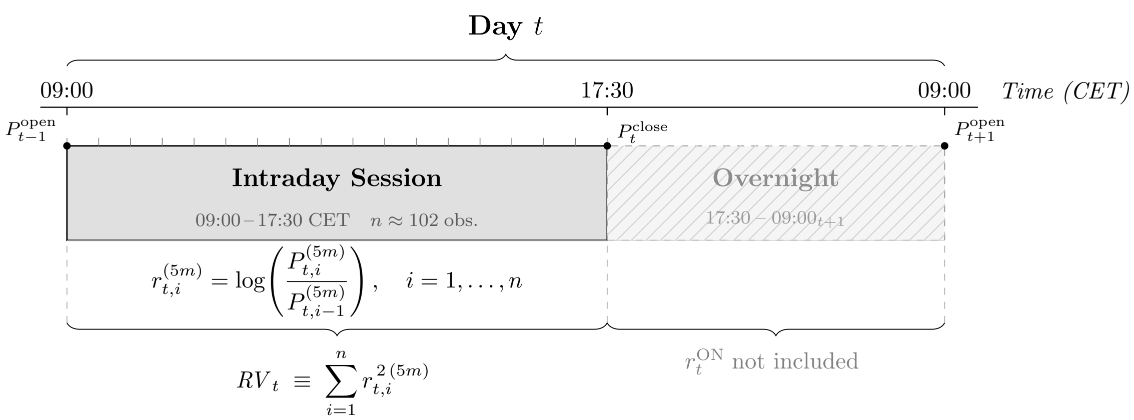
The final cleaned sample spans 5,489 trading days and encompasses several distinct macroeconomic regimes, including the Dot-com Bubble (2000 - 2002), the Global Financial Crisis (GFC, 2008 - 2009), the European Sovereign Debt Crisis (Euro Crisis, 2010 - 2012), a prolonged low-volatility expansion period (2016 - 2019), and the Covid-19 market turmoil of March 2020. This broad scope of market conditions makes the EURO STOXX 50 an especially demanding and informative testing ground for volatility forecasting models, as it allows us to evaluate model performance both in tranquil periods and during episodes of profound market uncertainty.

3.1 Trading Window

Official sources state that the EURO STOXX 50 is open from 09:00 to 17:30 CET, see STOXX Ltd. (2025). We perform diagnostic checks on our trading data and observe that, while there is a significant drop in recorded trading data post the 17:30 mark, trading data is still reported. However, this may stem from various factors including, but not limited to, closing auction mechanics, post-close index updates, sporadic recalculations, and vendor artificats. Because the data is structurally different outside of the regular trading window, we proceed to only focus our research on the intraday trading window from 09:00 to 17:30. This approach is consistent with adjacent research on the topic, including Christensen et al. (2023).

Moreover, while our trading records are reported in 1-minute intervals, we follow the standard approach in volatility forecasting literature which implies constructing 5-minute interval log-returns before proceeding with the realized variance computation. The underlying reason for this adjustments stems from the fact that 1-minute intervals can contain microstructure noise, such as order imbalances, order cancellations, and trading, and volatility forecasting has thus been shown to be improved by computing 5-minute intervals instead, see Bollerslev et al. (2016b). In Figure (1) we illustrate this composition in more detail.

Figure 1: Intraday Trading Window and Realized Variance Construction



Notes: The figure illustrates the daily sampling window used to construct the realized variance (RV) estimator for the EURO STOXX 50 index. The intraday session spans 09:00–17:30 CET, yielding $n = 102$ non-overlapping 5-minute log-returns per trading day. Using 5-minute intervals as opposed to 1-minute intervals is motivated to mitigate the adverse effects of market microstructure noise that inflate higher-frequency estimators. The overnight return from 17:30 to 09:00$_{t+1}$ CET is excluded from the estimator, as it is driven predominantly by non-trading-hours noise rather than the continuous-time price process; this restriction is standard in the volatility forecasting literature The daily realized variance in our case is therefore defined as the sum of squared 5-minute log returns.

Table 1: Training, Validation, and Test Set Partition

	Full Sample	**Training**	**Validation**	**Test**
Share of Sample	100%	70%	10%	20%
Observations	5,489	3,842	548	1,099
Start Date	1999-01-05	1999-01-05	2014-02-27	2016-04-21
End Date	2020-08-06	2014-02-26	2016-04-20	2020-08-06

Notes: The table reports the partition of the cleaned sample (5,489 trading days) into training, validation, and test sets. The training set is used for parameter estimation, the validation set for hyperparameter selection in the regularized linear models, gradient boosting, and neural network specifications, and the test set for out-of-sample forecast evaluation. For the non-regularized HAR benchmark models, bagging, and random forests, the training and validation sets are merged and parameters are re-estimated daily via a rolling window, consistent with Christensen et al. (2023).

3.2 Training-Validation-Test Split

In Table (1), we present the the sequential time-series split for the training (70%), validation (10%), and test (20%) sets respectively. The percentage split for each partition follows the broad set of literature on forecast modeling, including Christensen et al. (2023). Once we complete the data cleaning process, we obtain a total sample of 5,489 trading day observations spanning the period from the 5th of January 1999 to the 6th of August 2020.

3.3 Feature Sets

In volatility forecasting, the selection of explanatory variables plays an important role in determining predictive performance. Albeit the traditional HAR framework relies on a small set of realized measures that capture heterogenous persistence of volatility across different time horizons, novel machine learning techniques have allowed researchers and practioners to incorporate additional financial and macroeconomic variables to improve forecasting performance.

To examine how the information set affects forecasting performance, this study follows the approach of Christensen et al. (2023) and construct two different feature sets. The first feature set, denoted $\mathcal{M}_{\text{HAR}}$, contains only the core variables of the HAR framework, namely daily, weekly, and monthly monthly trailing realized variance components. All respective models forecasting performance are first evaluate using only this data set. This provides an apples-to-apples comparison when comparing models based on the same variables and data.

The second feature set, denoted $\mathcal{M}_{\text{ALL}}$, extends the baseline $\mathcal{M}_{\text{HAR}}$ feature set by incorporating a broader set of financial and macroeconomic predictors important in the European context to forecast volatility. By expanding the feature set, this specification allows novel machine learning models to exploit nonlinear relationships and interactions among a larger number of predictors.

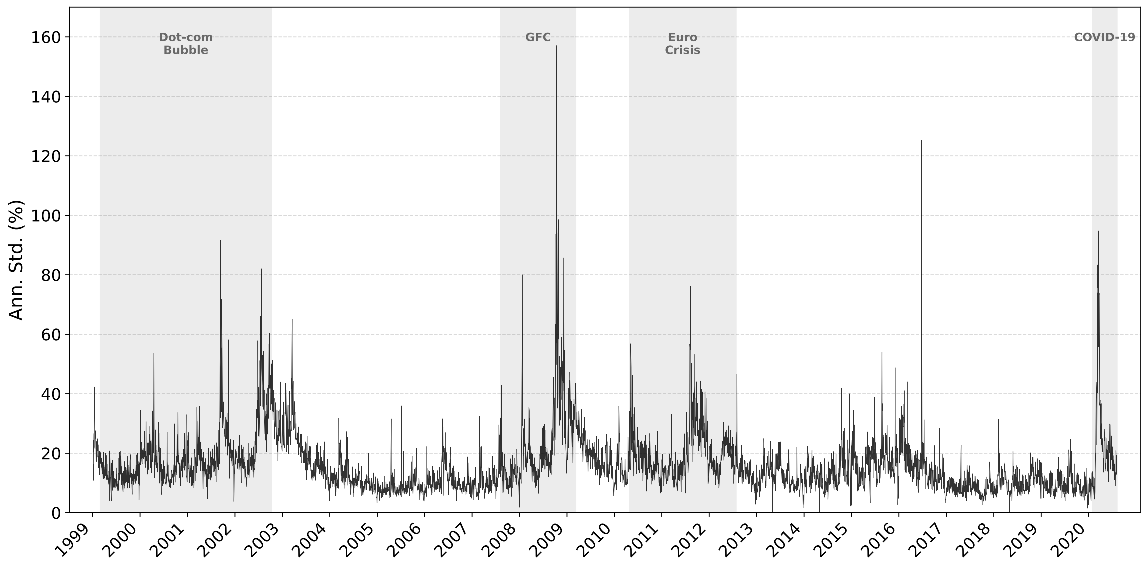
Comparing the forecasting performance across these two feature sets makes it possible to evaluate whether improvements in predictive accuracy primarily stem from the use of more flexible modeling techniques or from the inclusion of additional informative predictors.

3.3.1 $\mathcal{M}_{\text{HAR}}$

As previously noted, the $\mathcal{M}_{\text{HAR}}$ feature set is only composed of the trailing daily, weekly, and monthly realized variance compositions. The forecasting performance of all models presented in this paper are first evaluated using only this feature set. This approach will allow us to determine whether more complex and sophisticated machine learning techniques are superior compared to the conventional set of HAR models using the same predictors. However, it should be noted that the $\mathcal{M}_{\text{HAR}}$ feature set also includes variables that are specific to the various extensions of the HAR universe. For instance, the square-root of realized quarticity appears as an interaction term with realized variance in the HARQ model, but not in any of the other models.

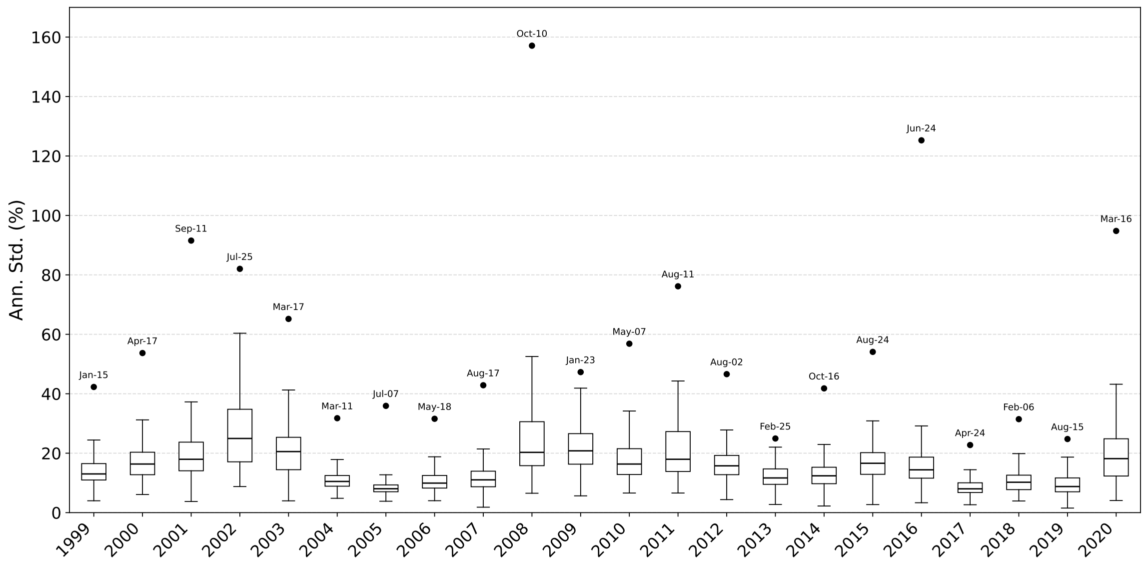
All data used to construct the $\mathcal{M}_{\text{HAR}}$ feature set are derived from the trading data set retrieved from the Stockholm School of Economics. For illustrational purposes, the distribution of annualized daily realized volatility, not variance, is depicted in Figures (**??**) and (3) over the entire sample period.

Figure 2: Annualized Daily Realized Volatility (1999 - 2020)



Notes: The figure plots annualized daily realized volatility for the EURO STOXX 50 index over the sample period January 1999 - August 2020. Shaded regions indicate periods of elevated market stress identified as distinct volatility regimes: the Dot-com Bubble (March 1999 – October 2002), the Global Financial Crisis (GFC, August 2007 – March 2009), the European Sovereign Debt Crisis (Euro Crisis, April 2010 – July 2012), and the COVID-19 pandemic (February 2020 – August 2020). To annualize the numbers, we assume an average of 252 trading days per year.

Figure 3: Distribution of Annualized Daily Realized Volatility (1999 - 2020)



Notes: The figure illustrates a box plot of the annual distribution of daily realized volatility for the EURO STOXX 50 over the sample period January 1999 - August 2020. Each box displays the interquartile range of daily annualized realized volatility within a given year, with the horizontal line in the box denoting the median. The dot above each box marks the single highest volatility observation of the year, with the corresponding date indicated above. Note that the box for 2020 only includes data from January till August which marks the end of our sample period. To annualize the numbers, we assume an average of 252 trading days per year.

3.3.2 $\mathcal{M}_{\text{ALL}}$

While the $\mathcal{M}_{\text{HAR}}$ feature set is restricted to realized measures derived solely from high-frequency trading data, the $\mathcal{M}_{\text{ALL}}$ feature set expands the information set by incorporating a broader range of financial and macroeconomic predictors. In contrast to $\mathcal{M}_{\text{HAR}}$, which captures the endogenous persistence of volatility through its heterogenous autoregressive structure, $\mathcal{M}_{\text{ALL}}$ integrates exogenous sources for information that reflect forward-looking expectations, macroeconomic conditions, commodity price dynamics, and global market spillovers. According to Christensen et al. (2023), such an enriched feature set allows the models to exploit a wider set of economic signals and potential nonlinear interactions, thereby providing a more comprehensive framework for forecasting volatility.

1. Endogenous Volatility and Risk Expectations

The VSTOXX index, maintained by Eurex and computed by Deutsche Börse, measures the implied volatility of the EURO STOXX 50 index derived from options prices with a 30-day horizon. As a forward-looking measure of market-implied uncertainty, the VSTOXX index captures the risk premium that investors demand for bearing volatility risk, and is thus commonly used to measure the level of investor fear in European equity markets. The inclusion of the VSTOXX is motivated by the well-documented leverage effect and volatility feedback channel, whereby rising implied volatility tends to predict elevated realized volatility in subsequent periods, see Blair et al. (2001) and Bekaert and Wu (2000). Furthermore, Christensen et al. (2023) show that implied volatility subsume much of the information contained in historical realized variance for forecasting purposes. We include both the level VSTOXX_t and its logarithm $\log(\text{VSTOXX}_t)$ to accomodate the well-known right-skewness of implied volatility. Daily VSTOXX index data was obtained from the STOXX website.

*2. **Macroeconomic and Systemic Risk Indicators***

Similarly to Christensen et al. (2023), we adopt the Economic Policy Uncertainty (EPU) index originally constructed by Baker et al. (2016). The index is constructed by counting the frequency of articles in major European newspapers that simultaneously contain terms related to the economic uncertainty and policies prevailing in the Eurozone. These counts are normalized by the total number of articles to ensure comparability over time and across sources. Note that the index is computed on a monthly basis, and we therefor compute a one month lagged version of this index to prevent look-ahead bias. The data is retrieved from the Economic Policy Uncertainty website.

In addition to the EPU index, we extend our paper by including the Composite Indicator of Systemic Stress (CISS) index developed Hollo et al. (2012) and provided by the European Central Bank (ECB). The daily index measures systemic risk in the European financial system by aggregating stress signals across multiple market segments, including equity, bond, money, and foreign exchange markets, as well as financial intermediaries. The CISS index is constructed from a set of market-based indicators such as volatility measures, credit spreads, liquidity indicators, which are transformed into standardized stress measures. We extract the daily CISS index from the ECB website.

*3. **Exchange Rate Dynamics***

We further extend our paper upon the contribution of Christensen et al. (2023) by accounting for global exchange rate dynamics with respect to the daily EUR/USD exchange rate. Fluctuations in the EUR/USD exchange rate reflects shift in relative monetary policy, risk sentiment, and international capital flows, all of which can influence the valuation and risk profile of the export-oriented international firms spanning the EURO STOXX 50 index. Empirically, the growing body of literature documents significant spillovers between foreign exchange and equity markets. For instance, Hau and Rey (2006) prove that exchange rates are strongly related to international equity flows and portfolio rebalancing. Moreover, Ülkü and Demirci (2012) investigate emerging markets in Europe and find that the sign of a country's relationship between between the exchange rate and domestic stock markets depend on a country's reliance of foreign capital as well as the level of liquidity in its equity markets. These findings support the inclusion of the EUR/USD exchange rate as an additional predictor in our extended feature set.

*4. **Commodity and Inflation-Linked Factor***

Contrary to Christensen et al. (2023), who focus on DJIA constituents in the US market, our European setting motivates the inclusion of the Brent crude oil price as an additional predictor. This decision stems from Europe's status as a net energy importer and the significant energy sector weight in the EURO STOXX 50 index, which increases the sensitivity of equity markets to energy price fluctuations. As a key global benchmark, the Brent crude oil price captures commodity price shocks that are closely linked to inflationary pressures and macroeconomic uncertainty, see Jawadi et al. (2023) for further details. The daily European Brent crude oil price is obtained from the US Energy Information Administration (EIA).

5. Global Market Spillovers

To account for global spillover effects, we extend the specification of Christensen et al. (2023) by including both Asian and US market spillovers, motivated by the position of the European markets in the global trading day. To account for Asian market spillovers, we retrieve daily returns from the Nikkei 225 index from the Federal Reserve Economic Data (FRED) website. Note that these daily returns are predetermined relative to the opening of the European markets. All days corresponding to Japanese holidays, where there is now trading activity, we proceed to forward-fill this gap.

Alongside the Nikkei 225 index, we retrieve daily returns for the S&P500 from the Wallstreet Journal Markets website. Note that contrary to the Asian markets, daily returns from the S&P500 are not predetermined relative to the opening of the European markets. Therefore, we lag the daily returns from the S&P500 by one day. Similarly to the approach for the Nikkei 225 index, we forward-fill all dates corresponding to American holidays where no trading takes place.

6. Momentum

Finally, we also include and measure the weekly and monthly momentum of the EURO STOXX 50 index in our $\mathcal{M}_{\text{ALL}}$ feature set. While Christensen et al. (2023) only incorporates a weekly momentum variable to measure financial performance of each of the DJIA constituents in their study, we compute both weekly ($M1W$) and monthly ($M1M$) momentum factors or our single-index setting. Momentum is constructed from daily close prices as 5-day and 22-day cumulative log-price changes.

3.4 Summary Statistics

In the following section, we present summary statistics for all variables encompassed by the $\mathcal{M}_{\text{HAR}}$ and $\mathcal{M}_{\text{ALL}}$ feature sets respectively for the entire sample period spanning from the 5th of January 1999 to the 6th of August 2020. The summary statistics are presented in Table (2). Further information, particularly pertaining to the evolution of the macroeconomic predictors in the $\mathcal{M}_{\text{ALL}}$ feature set, are provided in Appendix (8.3).

Table 2: Summary Statistics for the $\mathcal{M}_{\text{HAR}}$ and $\mathcal{M}_{\text{ALL}}$ Feature Sets

	Mean	**Std. Dev.**	**Skew.**	**Exc. Kurt.**	**Min**	**Median**	**Max**
Panel A: $\mathcal{M}_{\text{HAR}}$ Feature Set							
RVD	16.14	9.96	3.17	20.33	0.15	13.76	157.11
RVW	16.51	9.33	2.62	11.04	4.52	14.18	93.58
RVM	16.83	8.75	2.15	6.46	5.29	14.40	71.78
RVD^{+}	11.22	6.79	2.77	13.51	0.09	9.54	83.33
RVD^{-}	11.37	7.66	3.80	32.94	0.12	9.50	138.64
RD^{-}	−0.44	0.80	−3.37	18.27	−10.54	0.00	0.00
RW^{-}	−0.21	0.38	−3.83	25.44	−5.41	0.00	0.00
RM^{-}	−0.10	0.19	−3.51	17.55	−1.91	0.00	0.00
RQ	0.65	19.50	50.39	2,681.48	0.00	0.01	1,129.75
Panel B: Additional Variables Added to Create $\mathcal{M}_{\text{ALL}}$ Feature Set							
$VSTOXX$	23.91	9.78	1.71	4.18	10.68	21.98	87.51
EUR/USD_{ret}	0.00	0.61	0.01	2.93	−4.74	0.00	4.20
$BRENT_{ret}$	0.03	2.68	−2.04	81.37	−64.37	0.03	41.20
EPU	4.93	0.45	0.00	−0.90	4.07	4.99	6.07
$CISS$	0.17	0.19	1.76	3.30	0.00	0.11	0.94
$NIKKEI_{ret}$	0.01	1.44	−0.38	6.81	−12.11	0.00	13.23
$S\&P500_{ret}$	0.01	1.25	−0.37	10.72	−12.77	0.06	10.96
$NIKKEI_{sq}$	2.09	6.19	12.73	248.63	0.00	0.49	175.15
$S\&P500_{sq}$	1.56	5.56	13.49	263.52	0.00	0.31	162.95
$M1W$ (%)	−0.00	3.12	−0.75	5.54	−27.07	0.24	16.47
$M1M$ (%)	0.02	6.20	−1.22	5.20	−47.60	0.76	22.34

Notes: This table reports summary statistics for the full sample covering the period from 5th of January 1999 to the 6th of August 2020 for both the $\mathcal{M}_{\text{HAR}}$ and the $\mathcal{M}_{\text{ALL}}$ feature sets respectively. The variables RVD, RVW, RVM, RVD^{+}, and RVD^{-} are shown as annualized volatility for interpretation purposes. Moreover, the variables RD^{-}, RW^{-}, RM^{-}, EUR/USD_{ret}, $NIKKEI_{ret}$, $S\&P_{ret}$, $BRENT_{ret}$, $M1W$, and $M1M$ are shown as percentages. RQ is scaled by 10^6 while $NIKKEI_{sq}$ and $S\&P_{sq}$ are scaled by 10^4. The VSTOXX and CISS are shown in levels, and the EPU index is shown as the one month lagged log value of the index.

4 Methodology and Empirical Design

In this section, we outline the methodological framework and empirical design employed to generate the findings of our paper. We begin by introducing the concepts of realized variance and quarticity, after which we present the methodological structure underlying each forecasting model category and individual model.

4.1 Realized Variance and Quarticity

Let P_t denote the asset price. Following the literature on realized variance (Bollerslev et al. (2016b)), we view log-prices as evolving to the continuous-time diffusion depicted in Equation (1).

$$d\log(P_t) = \mu_t\,dt + \sigma_t\,dW_t \tag{1}$$

In the equation above, μ_t is the long-term trend that can possibly vary with time, σ_t is the instantaneous volatility process, and W_t is a standard Brownian motion. The object of interest is the latent *integrated variance* over day t, see Equation (2). This variable represents the cumulative variance generated by the continuous component of returns during the day.

$$IV_t = \int_{t-1}^{t} \sigma_s^2\,ds \tag{2}$$

In reality, however, IV_t is not directly observable. We therefore estimate the integrated variance term using high-frequency intraday trading data. We partition each trading day into n equally spaced intervals of length Δ (so that $n = 1/\Delta$). Albeit we do have access to 1-minute high-frequency trading data, we follow the conventional strand of literature on high-frequency data motivation of and proceed to construct 5-min high frequency intervals to avoid microstructure noise, see e.g. Li and Xiu (2016). As a result, we define intraday log-returns according to Equation (3).

$$r_{t,i}^{(5m)} = \log\left(\frac{P_{t-1+i\Delta}^{(5m)}}{P_{t-1+(i-1)\Delta}^{(5m)}}\right), \quad i = 1, \ldots, n \tag{3}$$

The proxy for integrated variance (IV_t), *realized variance*, can then be defined as the sum of squared intraday returns following Equation (4).

$$RV_t \equiv \sum_{i=1}^{n} r_{t,i}^{2\,(5m)} \tag{4}$$

Under standard regularity conditions and as $\Delta \to 0$, that is $n \to \infty$, realized variance is a consistent estimator of the latent integrated variance. In finite samples, however, realized variance is subject to non-trivial estimation error. In particular, the high-frequency limit theory implies

is a consistent estimator of IV_t.

In finite samples, however, realized variance is subject to non-trivial estimation error. In particular, the high-frequency limit theory implies the measurement-error representation

$$RV_t = IV_t + \eta_t, \qquad \eta_t \mid IQ_t \sim \mathcal{N}(0,\ 2\Delta\, IQ_t)\,, \tag{5}$$

where the error variance depends on the (latent) *integrated quarticity*

$$IQ_t = \int_{t-1}^{t} \sigma_s^4\, ds. \tag{6}$$

Intuitively, IQ_t scales the uncertainty in RV_t: days with larger IQ_t imply a noisier estimate of IV_t.

Because IQ_t is unobserved, we approximate it with *realized quarticity*,

$$RQ_t \equiv \frac{M}{3} \sum_{i=1}^{M} r_{t,i}^4, \tag{7}$$

which is consistent for IQ_t as the sampling frequency increases. Thus, RQ_t provides a feasible, time-varying proxy for the magnitude of the realized-variance measurement error and is central for distinguishing between days when RV_t is measured precisely versus imprecisely.

4.2 Benchmark Models

The benchmark models in our study comprise of the HAR model, developed by (Corsi (2009)), and its extension; LogHAR (Corsi (2009)), LevHAR (Corsi and Renò (2012)), SHAR (Patton and Sheppard (2015)), and HARQ (Bollerslev et al. (2016a)). Each model is described in more detail in the following subsections. The general linear equation for the standard HAR model and its extensions follows the structure of Equation (8), where Z_t denotes the feature set used to predict next day realized variance RV_{t+1}. The predictor vector $Z_t \in \mathbb{R}^p$ can thus be composed of the $\mathcal{M}_{\text{HAR}}$ or $\mathcal{M}_{\text{ALL}}$ feature sets respectively, as well as including the HAR-extension specific variables explained further in the sections below.

$$RV_{t+1} = \beta_0 + \beta' Z_t + \varepsilon_{t+1} \tag{8}$$

4.2.1 HAR Model

The Heterogeneous Autoregressive (HAR) model is a prasimonious time-series model designed to capture the long-term memory behavior of realized volatility usng a simple linear regression framework. It was introduced by Corsi (2009) to model high-frequency based volatility measures such as realized variance. The model is motivated by the Heterogenous Market Hypothesis, which argues that financial markets consist of agents operating at different investment horizons, such as short-term traders, medium-term portfolio managers, and long-term institutional investors.

Because these groups react to information over different time scales, volatility exhibits persistence across daily, weekly, and monthly horizons. The HAR model captures this multi-scale structure by explicitly including volatility components measured at different aggregation levels, thereby providing a simple approximation to long-memory dynamics.

$$RV_t = \beta_0 + \beta_d RV_{t-1} + \beta_w \overline{RV}^{(w)}_{t-1} + \beta_m \overline{RV}^{(m)}_{t-1} + \varepsilon_t \tag{9}$$

Here, RV_{t-1} is simply yesterday's realized variance while $\overline{RV}^{(w)}_{t-1}$ and $\overline{RV}^{(m)}_{t-1}$ denote the lagged weekly and monthly averages of realized variance, respectively. These components are constructed as rolling means over the previous 5 and 22 trading days, thereby excluding non-trading days.

$$\overline{RV}^{(w)}_{t-1} = \frac{1}{5}\sum_{i=1}^{5} RV_{t-i} \quad \text{and} \quad \overline{RV}^{(m)}_{t-1} = \frac{1}{22}\sum_{i=1}^{22} RV_{t-i} \tag{10}$$

The HAR model is often considered the baseline specification in the realized volatility literature, from which many extensions later have been developed.

4.2.2 LogHAR Model

The LogHAR model is simply a log-transformed version of the original HAR specification, estimated in logarithms rather than levels. The LogHAR model is typically applied because realized variance is strictly positive, highly right-skewed, and exhibits strong heteroskedasticity. Estimating the model in logarithms stabilizes the variance of the series, reducing the influence of extreme volatility spikes.

$$\log(RV_t) = \beta_0 + \beta_d \log\left(\overline{RV}_{t-1}\right) + \beta_w \log\left(\overline{RV}^{(w)}_{t-1}\right) + \beta_m \log\left(\overline{RV}^{(m)}_{t-1}\right) + \varepsilon_t \tag{11}$$

When the HAR model is estimated in logarithms, forecasts are produced for $log(RV_t)$. However, because the exponential function is nonlinear, simply exponentiating the predicted value leads to a downward-biased forecast of RV_t. To correct for this, we apply the standard log-normal bias adjustment.

$$\widehat{RV}_t = \exp\left(\widehat{\log(RV_t)} + \frac{1}{2}\widehat{\sigma}^2\right) \tag{12}$$

4.2.3 LevHAR Model

The standard HAR model captures volatility across horizons. However, it ignores an important empirical fact that negative returns today increases tomorrow's volatility more than positive returns of the same magnitude. This is known as the leverage effect. To account for this effect, Corsi and Renò (2012) extend the standard HAR model by incorporating aggregated negative returns, thereby introducing the so-called LevHAR model, see equation (13).

$$RV_t = \beta_0 + \beta_d RV_{t-1} + \beta_w \overline{RV}_{t-1}^{(w)} + \beta_m \overline{RV}_{t-1}^{(m)} + \underbrace{\gamma_d r_{t-1}^{-} + \gamma_w \overline{r}_{t-1}^{(w)-} + \gamma_m \overline{r}_{t-1}^{(m)-}}_{\text{Leverage effect}} + \varepsilon_t \tag{13}$$

$$r_{t-i}^{-(h)} = \min\{0, r_{t-i}^{(h)}\} \quad \text{where} \quad h \in \{1, 5, 22\} \quad \text{and} \quad r_{t-1}^{(h)} = \frac{1}{h}\sum_{i=1}^{h} r_{t-i} \tag{14}$$

The additional terms in equation (13) capture the leverage effect, whereby negative returns are associated with subsequent increases in volatility. While the standard HAR model treats past volatility symmetrically, the LevHAR extension of including leverage effects in the standard HAR model assign additional weight to adverse market movements. This asymmetry is well documented in the volatility forecasting literature, where negative returns are found to have a stronger and more persistent impact on future volatility, whereas positive returns exhibit little so no predictive power, see Corsi (2009) and Corsi and Renò (2012).

4.2.4 SHAR Model

Closely related to the LevHAR model, the Semivariance HAR (SHAR) model introduced by Patton and Sheppard (2015) provides an alternative way of capturing asymmetry in volatility dynamics. While LevHAR incorporates negative returns directly to model the leverage effect, SHAR decomposes realized variance into its positive and negative semivariance components, allowing volatility to respond differently to upside and downside variation.

$$RV_t = \beta_0 + \beta_d^{+} RV_{t-1}^{+} + \beta_d^{-} RV_{t-1}^{-} + \beta_w \overline{RV}_{t-1}^{(w)} + \beta_m \overline{RV}_{t-1}^{(m)} + \varepsilon_t \tag{15}$$

$$RV_{t-1} \equiv RV_{t-1}^{+} + RV_{t-1}^{-} \tag{16}$$

In this specification, RV_{t-1}^{+} and RV_{t-1}^{-} denote the positive and negative realized semivariances, constructed as the sum of intraday and overnight squared returns conditional on returns being positive and negative, respectively. This decomposition allows the model to distinguish between upside and downside volatility, reflecting the empirical finding that negative price movements typically exert stronger and more persistence effect on future volatility than positive movements of similar magnitude.

4.2.5 HARQ Model

The HARQ model was introduced by Bollerslev et al. (2016a) and extends the standard HAR framework by explicitly accounting for the time-varying measurement error in realized variance. Since realized variance is a generated regressor, it contains estimation noise whose variance is proportional to integrated quarticity. Ignoring this induces an attenuation bias in the HAR coeffcients. The HARQ model corrects this errors-in-variables problem by allowing the autoregressive coefficient on daily realized variance to vary with the square troot of realize quarticity. When measurement error is small, persistence is higher; when measurement error is large, the influence of lagged realized variance is reduced. This adjustment leads to stronger average persistence and improved volatility forecasting.

$$RV_t = \beta_0 + \underbrace{\left(\beta_1 + \beta_{1RQ}\sqrt{RQ_{t-1}}\right)}_{\beta_{1,t}} RV_{t-1} + \beta_2 RV^{(w)}_{t-1} + \beta_3 RV^{(m)}_{t-1} + \varepsilon_t. \tag{17}$$

Here, RQ_{t-1} represents yesterday's realized quarticity which makes the coefficient $\beta_{1,t}$ vary with time. In essence, $\sqrt{RQ_{t-1}}$ proxies the standard deviation of the measurement error in RV_{t-1}. As argued by Bollerslev et al. (2016a), the HARQ adjustment is applied only to the daily lag because temporal aggregation in the weekly and monthly averages substantially reduces measurement error and, hence, attenuation bias, rendering additional Q-adjustments for those components largely unnecessary.

4.2.6 HAR-X Model

Finally, to round-off the section on the linear benchmark models, we also introduce the HAR-X model, similarly to Christensen et al. (2023). Note that that the HAR-X model is simply the standard HAR model, but under the $\mathcal{M}_{\text{ALL}}$ feature set the HAR-X model integrates all the additional macroeconomic predictors in the linear model. As such, the HAR-X model can be formulated according to equation (18), where X_{t-1} denotes the additional set of predictors under the $\mathcal{M}_{\text{ALL}}$ feature set and γ' represents the vector of coefficients.

$$RV_t = \beta_0 + \beta_d RV_{t-1} + \beta_w \overline{RV}^{(w)}_{t-1} + \beta_m \overline{RV}^{(m)}_{t-1} + \gamma' X_{t-1} + \varepsilon_t \tag{18}$$

4.3 Regularized Linear Models

Regularized linear models are a set of extensions of the standard linear regression (OLS) models that incorporates penalty term to control model complexity and prevent overfitting.

The general loss function for a regularized linear model can be written as:

$$\min_{\beta_0,\beta} \left\{ \sum_{t=1}^{T} (y_t - \beta_0 - X_t\beta)^2 + \lambda \cdot \phi(\beta) \right\} \tag{19}$$

In the setting above, the first component of the expression, the sum of squared residuals $\sum_{t=1}^{T}(y_t - \beta_0 - X_t\beta)^2$, corresponds to the standard ordinary least squares loss function and measures how well the model fits the observed data. In the context of this paper, y_t denotes the the realized variance at time t, white X_t represents the vector of predictors, such as lagged realized variance and additional explanatory variables included in the extended feature set $\mathcal{M}_{\text{ALL}}$. Minimizing this first term of the equation would yield the OLS estimator, which is optimal in-sample but prone to overfitting when the number of predictors is large or when multicolliniarity is present, both of prevalent in high-frequency volatility forecasting.

To address this issue, the model incorporates a penalty term $\lambda \cdot \phi(\beta)$, where $\phi(\beta)$ is a function of the regression coefficients and $\lambda \geq 0$ is a tuning parameter controlling the strength of regularization. Note that if $\lambda = 0$, the regularized linear model is the same as the standard linear regression model. This term thus imposes a cost on large coefficients, effectively shrinking them towards zero. The inclusion of this penalty introduces bias into the estimate but reduces their variance, with the objective to improve out-of-sample performance.

Different versions of $\phi(\beta)$ give rise to different regularization models. For instance, a quadratic penalty term corresponds to ridge regression (RR), an absolute value penalty terms leads to the LASSO (LA) regression, and a combination of both RR and LA yields the elastic net (EN). In the following subsections, we follow the approach of Christensen et al. (2023) and present these three regularized linear regression models.

4.3.1 Ridge Regression

The Ridge Regression (RR), originally crafted by Hoerl and Kennard (1970), aims to reduce variance of the model by adding a quadratic penalty term to shrinks all coefficients continuously toward zero, but never exactly to zero. This is known as the L_2 penalty term and thus penalizes large coefficients more heavily. The penalty term is detailed in equation 20.

$$\phi(\beta) = \sum_{j=1}^{p} \beta_j^2 \tag{20}$$

Adopting this technique can prove to be highly useful in improving volatility forecasting with feature sets where variables are highly correlated and the model risks overfitting noise.

4.3.2 LASSO Regression

The LASSO (LA) regression is another form of regularized linear model introduced by Tibshirani (1996). Unlike the ridge regression, the LA regression instead adopts what is called the L_1 penalty term to the original ordinary least squares regression, which sums the absolute value of the coefficients.

$$\phi(\beta) = \sum_{j=1}^{p} |\beta_j| \tag{21}$$

With this alternative set-up for the penalty term presented in equation (21), the loss function of the LA regression can effectively shrink coefficients to zero, effectively performing variable selection. This characteristic of the LA regression will prove efficient to identify the most relevant drivers of next-day volatility.

4.3.3 Elastic Net

The Elastic Net (EN) regression, derived by Zou and Hastie (2005), can be interpreted as a combination of the RR and LA regression, designed to overcome the individual shortcomings of each these models. While the RR retains all predictors but fails to perform variable selection, and the LA regression performs variable selection but struggle with stability when predictors are highly correlated, the EN combines both the L_1 and L_2 penalty terms respectively into a single penalty function. This penalty term is captured by equation (22).

$$\phi(\beta) = \alpha \sum_{j=1}^{p} |\beta_j| + (1-\alpha) \sum_{j=1}^{p} \beta_j^2 \tag{22}$$

In equation (22), $\alpha \in [0, 1]$ is the mixing parameter governing the contribution of each penalty. Assigning $\alpha = 1$ yields the LA model, while setting $\alpha = 0$ produces the RR model, such that the EN nests both models as special cases.

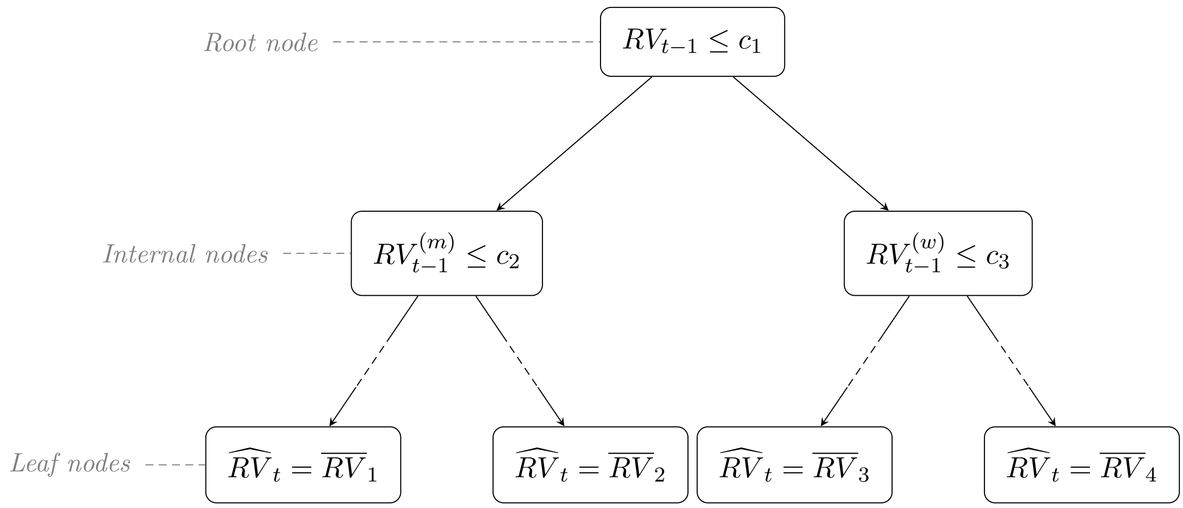
4.4 Tree-based Models

Tree-based models belong to another set of frameworks well-suited for volatility forecasting because the characteristics of these models take into account the various features of financial data, namely nonlinearity, interaction effects, and high-dimensional data. Therefore, tree-based models address many of the core limitations of conventional linear models.

A single decision tree is a non-parametric machine learning approach that makes predictions from the data by recursively splitting the data into subsets based on feature values and criteria, forming a tree-like structure of decision rules. Predictions are made by averaging the values that end up in the terminal nodes of the tree, known as leaf nodes. Figure (4) serves as an illustratory example of a single decision tree.

Similarly to the approach presented in Christensen et al. (2023), we apply three well-documented volatility forecasting tree-based models in our paper. These are; Bagging (BG) Breiman (1996), Random Forest (RF) Breiman (2001), and Gradient Boosting (GB) Friedman (2001).

Figure 4: Illustration of a Regression Tree for Realized Variance



Notes: The figure illustrates a stylized regression tree using $\mathcal{M}_{\text{HAR}}$ predictors. Each internal node partitions the predictor space based on threshold values c_1, c_2, c_3, and each leaf node reports the average realized variance $\overline{RV}_j$ within that region where j is the total number of leaf nodes in the tree. The dashed segments along the second-level edges indicate that additional splits may occur before reaching the leaf nodes.

4.4.1 Bagging

Single decision trees have a static structure, and because of this characteristic single decision trees are prone to overfitting, inherently unstable, and subject to high variance. To address these shortcomings, Breiman (1996) proposed a bootstrapped aggregation technique, whereby multiple independent trees are built and estimated on resampled subsets of the original data and subsequently averaged to make a prediction. All subsets of the data are equally large and randomly formed, including replacement. The predictions are derived from the BG model following equation (23).

$$\hat{f}(Z_t) = \frac{1}{B} \sum_{b=1}^{B} \hat{f}^{(b)}(Z_t) \tag{23}$$

In the equation above, $\hat{f}(Z_t)$ denotes the prediction from the BG model and $\hat{f}^{(b)}(Z_t)$ represents the prediction from the bth bootstrapped sample, i.e. the bth tree. Moreover, Z_t is the feature set and B is the total number of trees in the model. Note that all independent variables are included in all trees formed by applying the BG method.

4.4.2 Random Forest

Random Forests (RF), introduced by Breiman (2001), is a machine learning method designed to improve predictive out-of-sample performance by combining and averaging a large number of regression trees. The approach builds directly on the concept of bootstrap aggregation (bagging, BG). While BG reduces variance by averaging across independently estimated trees, RF extends this framework by introducing an additional layer of randomness through feature sub-sampling, with the objective to further decorrelate individual trees and enhance out-of-sample performance.

Let $Z_t \in \mathbb{R}^p$ be the feature set, i.e. the predictor vector, and y_t the target variable, which in this study corresponds to the actual one-day-ahead realized variance. The RF predictor is defined in equation (24).

$$\hat{f}(Z_t) = \frac{1}{B} \sum_{b=1}^{B} \hat{f}^{(b)}(Z_t, \theta^{(b)}) \tag{24}$$

In this equation, B denotes the number of trees used in the forest and $\hat{f}^{(b)}(\cdot)$ is the prediction stemming from the b-th regression tree. The only characteristic that make the RF model distinct from BG is the $\theta^{(b)}$ notation, which embodies the random set of feature variables used in tree b. Each tree partitions the predictor space into a set of disjoint regions, so called leaf nodes, and assigns a constant prediction within each region. Specifically, a regression tree can be written as illustrated in equation (25).

$$\hat{f}^{(b)}(Z_t) = \sum_{m=1}^{M_b} c_m^{(b)} \, \mathbf{1}\{Z_t \in R_m^{(b)}\} \tag{25}$$

Here, $\left\{R_m^{(b)}\right\}_{m=1}^{M_b}$ represents the partition of the feature space created by tree b, and $c_m^{(b)}$ is the average of the target variable within region $R_m^{(b)}$.

The constellation of the RF model relies on two critical sources of randomness. First, as previously alluded to, each tree is estimated on a bootstrap sample drawn with replacement from the original data set. Let $\mathcal{D} = \{(x_t, y_t)\}_{t=1}^{T}$. For each tree b, a bootstrap sample $\mathcal{D}^{(b)}$ is formed, introducing variability across trees and reducing the tendency of individual decision trees to overfit the data.

Second, and more importantly, RF differs from standard BG by introducing random feature sub-sampling at each split in the tree. Rather than cosidering all p predictors in the feature set Z_t when determining the optimal split, only a randomly selected subset of size m predictors is evaluated. In this paper, we follow the conventional recommendation for regression forests and define m according to equation (26).

$$m = \left\lfloor \frac{p}{3} \right\rfloor \tag{26}$$

This specification is well established in the literature on RF (Breiman (2001), Hastie

(2009)) and serves to reduce the correlation between trees. Intuitively, if all predictors were considered at each split, dominant explanatory variables, such as daily lagged realized variance, would be repeatedly selected across trees, resulting in highly similar and correlated tree structures. By restricting the candidate set of predictors for each tree b according to (26), the RF model forces individual trees to explore alternative splits, thereby increasing model diversity and improving the effectiveness of averaging.

For volatility forecasting, this behavior is particularly useful, as volatility dynamics are known to exhibit nonlinearities and complex interactions across time horizons and macro-financial variables, see Christensen et al. (2023). RF can flexibly capture such patterns without requiring explicit specification of interaction terms or functional forms.

Following the literature on the application of RF, we adopt the same approach as Christensen et al. (2023) and implement the RF model using a standard set of hyperparameters. The number of trees, B, is set to 500 and each tree is grown subject to a minimum leaf size constraint of 5 observations per leaf node, see Hastie (2009).

4.4.3 Gradient Boosting

Gradient Boosting (GB) is a tree-based machine learning algorithm originally proposed Friedman (2001). Rather than fitting a single complex model, GB constructs and ensemble of shallow decision trees sequentially, where each tree is trained to correct the errors of the preceding ensemble. This iterative refinement procedure is grounded in the principle of functional gradient descent, where the negative gradient of the loss function with respect to the current prediction serves as the target for each new tree.

Formally, the GB algorithm is initialized with a constant prediction as described by the equation below.

$$\hat{f}^{(0)}(Z_t) = \arg\min_{\beta_0} \sum_{t \in \text{train}} \left(RV_t - \beta_0\right)^2, \tag{27}$$

Under the mean squared error loss function, the model reduces the sample mean of realized variance. At each subsequent iteration $b = 1, \ldots, B$, a shallow regression tree fitted to the residuals of the current ensemble, and the prediction is updated according to equation (28).

$$\hat{f}^{(b)}(Z_t) = \hat{f}^{(b-1)}(Z_t) + \nu \sum_{j=1}^{J_b} \hat{\gamma}_{jb} \, \mathbf{1}_{\{Z_t \in R_{jb}\}}, \tag{28}$$

Here, R_{jb} denotes the j-th terminal node of the b-th tree, $\hat{\gamma}_{jb}$ is the optimal step size with that node, and $\nu \in (0, 1]$ is a learning rate that controls the contribution of each tree to the final prediction. The final forecast is provided by the accumulated ensemble $\hat{f}(Z_t) \equiv \hat{f}^{(B)}(Z_t)$.

Each individual tree in the ensemble constitutes a so-called weak learner, a model only marginally better than a naive benchmark, but their sequential combination produces powerful nonparametric forecasting model capable of capturing nonlinearities and interaction effects in the predictor space without requiring these to be specified by the researcher *a priori*.

A well-known limitation of GB in financial applications is its sensitivity to outliers. Because the algorithm places disproportionate weight on large residuals when fitting each successive tree, extreme observation, such as volatility spikes during market stress episodes, can distort the learned function. This stands in contrast to Random Forests, whcih mitigates this issue through independent bootstrap sampling. Given this pronounced right-skewness and heavy tails of realized variance, this sensitivity is a relevant consideration in our setting, as noted by Christensen et al. (2023).

Following Christensen et al. (2023), the GB model is estimated using a rolling window scheme, where each forecast origin the hyperparameters are re-selected on the validation portion of the current window. Specifically, the tree depth is tuned over $\{1, 2\}$, the learning rate over $\{0.01, 0.1\}$, and the number of trees over $\{50, 100, \ldots, 500\}$, with all remaining parameters set to their default values as proposed in the original implementation. The combination of minimizing the mean squared error on the validation set is retained, and the model is re-fitted on the full training window prior to generating the one-day-ahead forecast. Any negative forecast is replaced by zero, following standard practice in the realized variance literature.

4.5 Neural Networks

Neural networks (NNs) represent a class of machine learning algorithms inspired by the structure of the human brain. Unlike conventional simple linear models such as the HAR and its extensions, NNs are capable of approximating highly nonlinear and complex functional relationships between predictors and the response variable. This flexibility makes them particularly well-suited for financial time series, where the underlying data-generating process is known to be nonlinear, exhibits low signal-to-noise ratios, and is subject to structural breaks and regime changes over time. As noted by Christensen et al. (2023), the ability of NNs to capture nonlinearities in the dynamic structure of realized variance is a key source of forecast improvement relative to the linear HAR benchmark models.

4.5.1 Feed-forward Neural Architectures

We proceed to implement four different types of feed-forward neural networks in which information flows in a single direction from the input layer through one or more hidden layers to finally reach the output layer. We adopt the same four neural architectures as Christensen et al. (2023) and they are denoted as NN_1, NN_2, NN_3, and NN_4 respectively. See Figure (5) for details on the respective architecture of each neural network. Here, the input layer receives the either the $\mathcal{M}_{\mathrm{HAR}}$ or the $\mathcal{M}_{\mathrm{ALL}}$ feature set denoted as Z_t. The hidden layer builds a sequence of nonlinear transformations, and the output produces the prediction $\hat{f}(Z_t)$.

Let L represent the number of layers in a given NN architecture and let N_l denote the total number of neurons in layer l. Formally, the general equation for the l-th hidden layer is given by Equation (29).

$$a_t^{(\theta_{l+1},b_{l+1})} = g_l\left(\sum_{j=1}^{N_l} \theta_j^{(l)} a_t^{(\theta_l,b_l)} + b^{(l)}\right), \quad 1 \leq l \leq L \tag{29}$$

In this equation, $g_l(\cdot)$ denotes the activation function, $\theta^{(l)}$ is the weight matrix, $b^{(l)}$ is the bias vector serving as an activation threshold, and $a_t^{(\theta_{l+1},b_{l+1})}$ is the output of layer $l+1$. Note that $a_t^{(\theta_l,b_l)}$ serves as the input layer for layer $l+1$. The final forecast is constructed by composing the transformation across all layers, see Equation (30) where each layer adds a new level of transformation to the NN.

$$\hat{f}(Z_t) = (g_L \circ \cdots \circ g_1)(Z_t) \tag{30}$$

In accordance with Christensen et al. (2023), we adopt the Leaky Rectified Linear Unit (L-ReLU) of Maas et al. (2013) as the activation function for all hidden layers, see Equation (31).

$$\text{L-ReLU}(x) = \begin{cases} cx & \text{if } x < 0 \\ x & \text{otherwise} \end{cases} \tag{31}$$

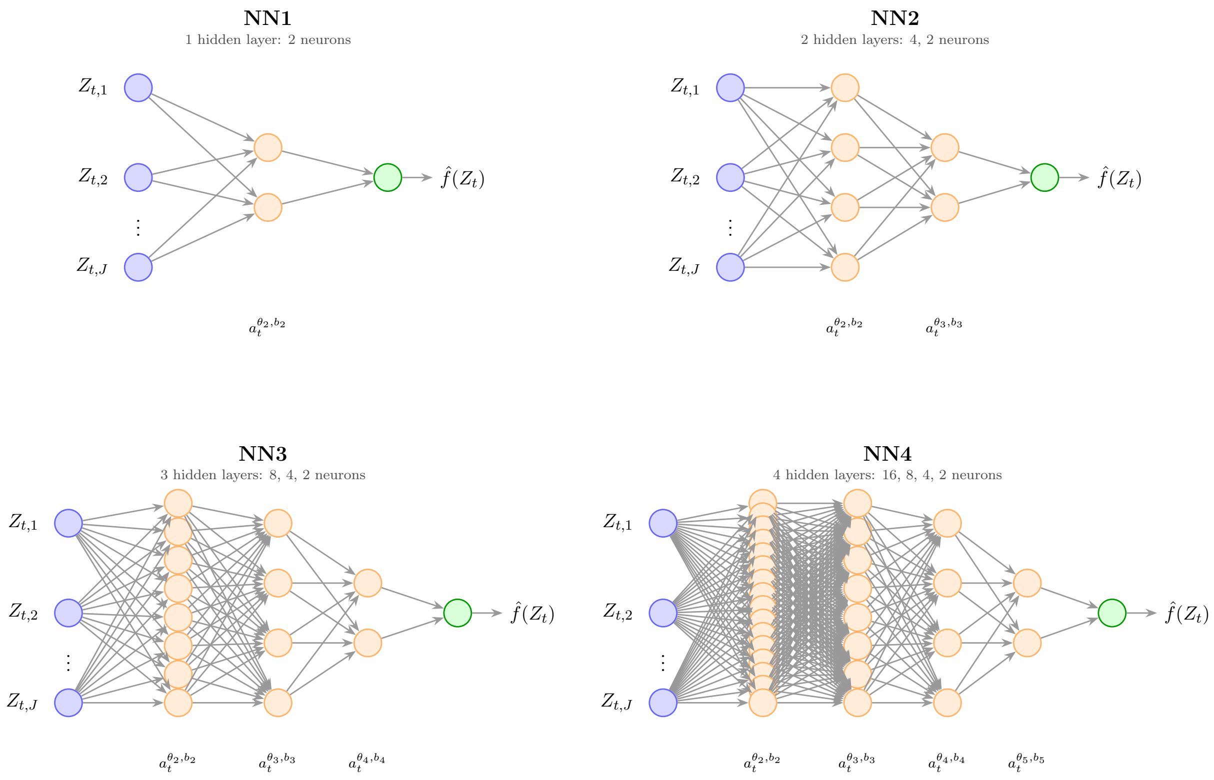
L-ReLU improves upon the standard ReLU activation function by allowing a small, non-zero slope, c, for negative inputs to prevent neurons from dying in the network. Following Christensen et al. (2023), we set c in Equation (31) equal to 0.01. The output layer uses a linear activation function, appropriate for a regression problem where the target variable is continuous.

In addition, we also add a drop-out feature to our NNs as a form of regularization technique. Drop-out attempts to prevent the NN from becoming to dependent on one or a few neurons, thereby serving the purpose of reducing overfitting, see Equation (32). It works by randomly deactivating neurons in a given layer l during training.

$$\tilde{a}_t^{(\theta_l,b_l)} = d^{(\theta_l,b_l)} \odot a_t^{(\theta_l,b_l)} \quad \text{where} \quad d_i^{(\theta_l,b_l)} \sim \text{Bernoulli}(p) \tag{32}$$

In this equation, $d^{(\theta_l,b_l)}$ is a masked vector performing the drop-out through element-wise multiplication with layer $a_t^{(\theta_l,b_l)}$. We set $p = 0.8$, implying that we keep 80% of the neurons in any given layer l.

Figure 5: Feed-Forward Neural Network Architectures: NN1, NN2, NN3, and NN4



Notes: The figure illustrates the four feed-forward neural network architectures considered in this study, following Christensen et al. (2023) and inspired by the geometric pyramid structure of **?** and **?**. Each network receives J input features $Z_{t,1}, \ldots, Z_{t,J}$, where $J = 3$ for the $\mathcal{M}_{\text{HAR}}$ feature set and $J = 20$ for the extended $\mathcal{M}_{\text{ALL}}$ feature set. NN1 is a shallow network with a single hidden layer of 2 neurons; NN2 has two hidden layers of 4 and 2 neurons; NN3 has three hidden layers of 8, 4, and 2 neurons; NN4 has four hidden layers of 16, 8, 4, and 2 neurons. All hidden layers apply a Leaky ReLU activation function with negative slope $c = 0.01$ and are subject to dropout regularization with a keep rate of 0.8. The output layer returns the one-step-ahead forecast $\hat{f}(Z_t)$ via a linear activation function.

4.6 Out-of-Sample Performance Evaluation

To assess the predictive performance and accuracy of the different forecasting models on each of the two feature sets $\mathcal{M}_{\text{HAR}}$ and $\mathcal{M}_{\text{ALL}}$, we employ a set of different out-of-sample evaluation measures. Following the same methodological approach as Christensen et al. (2023), our primary loss function is the Mean Squared Error (MSE). We supplement this performance measure by computing a pairwise tests of equal predictive accuracy using the framework introduced by Diebold and Mariano (2002).

4.6.1 Relative Mean Squared Error

Let $\hat{f}(Z_t)$ denote the one-day-ahead forecast of realized variance (RV_{t+1}) provided by one of the forecasting models at time t, based on the feature set Z_t. The feature set Z_t will be either the $\mathcal{M}_{\text{HAR}}$ or the $\mathcal{M}_{\text{ALL}}$ feature set respectively. The MSE of the model is defined in equation (33).

$$\mathcal{L}_{MSE}(\hat{f}(Z_t)) = \frac{1}{T} \sum_{t \in \mathrm{OOS}} \left(RV_{t+1} - \hat{f}(Z_t) \right)^2 \tag{33}$$

In this equation, the out-of-sample denotes the number of observations, i.e. trading days, in our test sample, and RV_{t+1} is the actual realized variance on trading day $t+1$. Since the MSE performance measure serves as the primary loss function underlying the estimation of all models in our study, it provides a natural and consistent basis for out-of-sample performance evaluation, see Christensen et al. (2023).

To facilitate model comparison, we report a pairwise *relative MSE* score for each model pair (i, j), defined in equation (34).

$$\text{Relative MSE}_{i,j} = \frac{\text{MSE}_i}{\text{MSE}_j} \tag{34}$$

Here, model j serves as the benchmark. A ratio below 1 indicates that model i outperforms the benchmark model j, while a ratio above 1 implies the opposites. For our results, see Section 5, we report a relative MSE score between each model pair in a more comprehensive format , see Tables (3). For ease of interpretation, we also illustrate the relative MSE och each model relative to the benchmark HAR model of Corsi (2009), see Figures (**??**) since this is one of the most widely used specifications in the realized volatility forecasting literature.

4.6.2 Diebold-Mariano Test

While the relative MSE quantifies the magnitude of the forecasting accuracy of a model pair, it does not provide a format assessment of whether the difference in forecasting accuracy is statistically significant. We therefore complement the relative MSE with the Diebold and Mariano (2002) test for equal predictive accuracy. For a given model pair (i, j), the loss differential series is defined in equation (35).

$$d_t = \left(RV_{t+1} - \hat{f}_i(Z_t) \right)^2 - \left(RV_{t+1} - \hat{f}_j(Z_t) \right)^2 \tag{35}$$

In this specification, a negative value of d_t suggests that model i produces a smaller squared error that model j at time t. Moreover, the null hypothesis of equal predictive accuracy is provided in equation (36) against the one-sided alternative in equation (37)

$$H_0 : \mathbb{E}[d_t] = 0 \tag{36}$$

$$H_1 : \mathbb{E}[d_t] > 0 \tag{37}$$

The alternative hypothesis H_1 suggests that model j outperforms model i The Diebold and Mariano (2002) test statistic is provided by computing equation (38).

$$DM = \frac{\bar{d}}{\sqrt{\widehat{\text{Var}}(\bar{d})}} \tag{38}$$

Here, $\bar{d}$ is the sample mean of the loss differential series, and $\widehat{\text{Var}}(\bar{d})$ is a heteroskedasticity and autocorrelation consistent (HAC) variance estimator to account for potential serial correlation in d_t. Under the null hypothesis of equal predictive accuracy, i.e. that none of the models outperforms the other, the test statistic asymptotically follows a standard normal distribution, see equation (39).

$$DM \xrightarrow{d} \mathcal{N}(0,1) \tag{39}$$

4.6.3 Additional Loss Functions

While the MSE serves as the primary loss function in our empirical analysis of the results, alternative loss functions may provide complementary insights to evaluate model performance. Therefore, apart from the relative MSE and the pairwise Diebold and Mariano (2002) test, we also report the Mean Absolute Error (MAE) and the Root Mean Squared Error (RMSE) for each model across both feature sets.

The MAE loss function reports the average absolute deviation between the actual one-day-ahead realized variance (RV_{t+1}) and the forecasted one-day-ahead realized variance ($\hat{f}(Z_t)$) for each model. Contrary to the MSE, which squares the forecast errors, the MAE loss function imposes a linear penalty on deviations, thereby making the performance evaluation less sensitive to large forecasting errors. As such, the MAE can be particularly useful in the context of volatility forecasting, where extreme volatility movements, such as those observed during the market turmoil amidst the Covid-19 pandemic, may disproportionally benefit some models. The MAE loss function is presented in equation (40).

$$\mathcal{L}_{MAE}(\hat{f}(Z_t)) = \frac{1}{T} \sum_{t \in \text{OOS}} \left| RV_{t+1} - \hat{f}(Z_t) \right| \tag{40}$$

The RMSE loss function is closely related to the MSE. By taking the root of the MSE, the RMSE rescales the loss function to the original units of the dependent variable realized variance. Therefore, the RMSE retains the qualities of the quadratic penalization of the MSE while facilitating a more direct economic interpretation of the resulting loss value due to unit consistency. The RMSE is constructed in equation (41).

$$\mathcal{L}_{RMSE}(\hat{f}(Z_t)) = \sqrt{\frac{1}{T} \sum_{t \in \text{OOS}} \left(RV_{t+1} - \hat{f}(Z_t) \right)^2} \tag{41}$$

5 Empirical Results

In the following sections, we present the empirical findings of our study. We begin by presenting the performance of each model using only the predictors available in the $\mathcal{M}_{\text{HAR}}$ feature set. We then proceed to compare these results with the results obtained from allowing the more advanced ML models to exploit nonlinearities and more complex patterns from the additional variables available in the extensive $\mathcal{M}_{\text{HAR}}$ feature set.

5.1 Results from $\mathcal{M}_{\text{HAR}}$ Feature Set

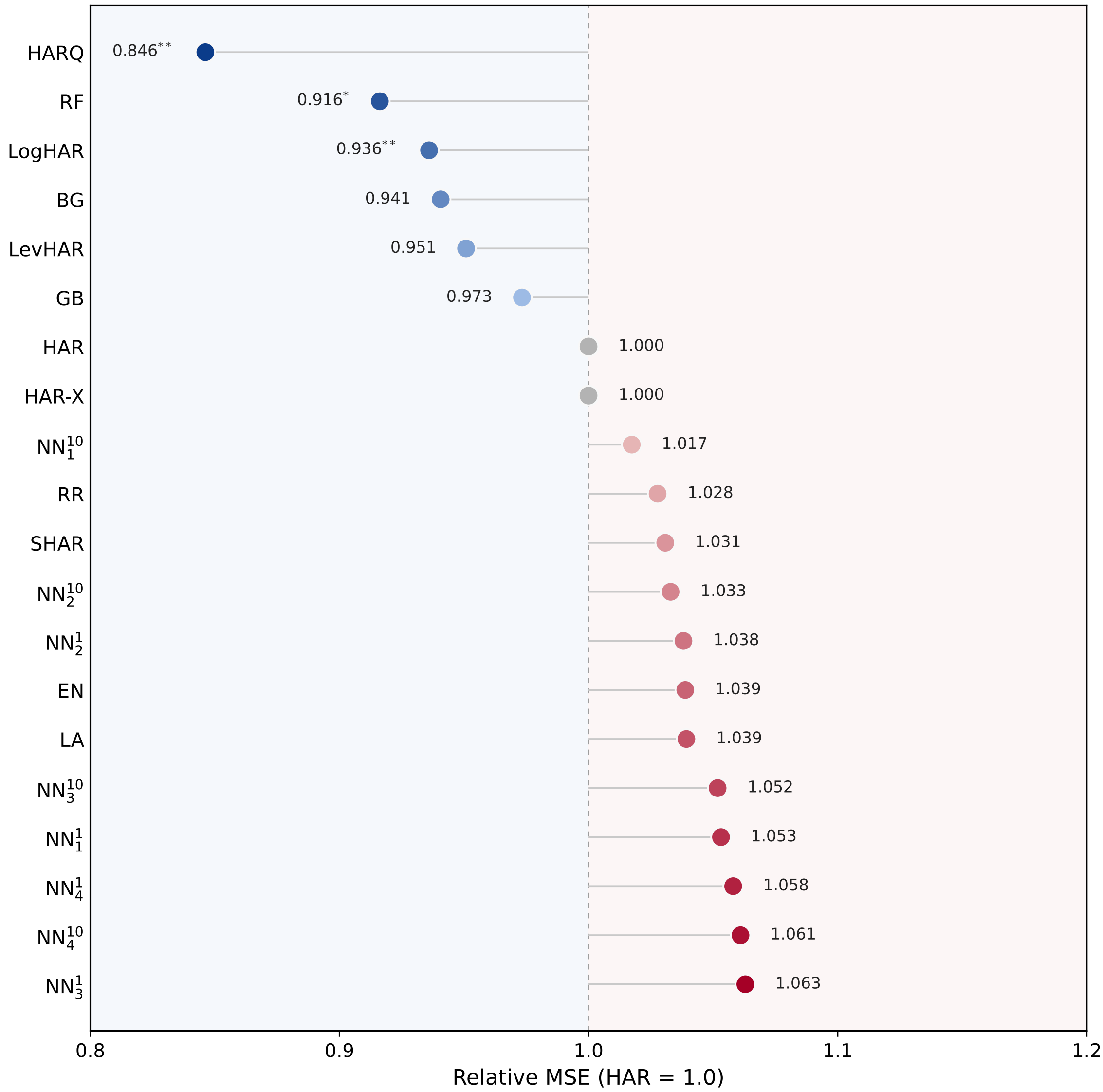
In this section, we present our empirical findings for $\mathcal{M}_{\text{HAR}}$ feature set. Table (3) illustrates the pairwise relative MSE performance of each model pair. Moreover, we conduct the Diebold and Mariano (2002) test to observe whether the difference in model performances is statistically significant or not. In Figure (6) we illustrate the relative MSE relative to the standard HAR model, with ranked performance from best to worst model for ease of interpretation.

Table 3: Pairwise Relative MSE and Diebold-Mariano Tests on $\mathcal{M}_{\text{HAR}}$ Feature Set

	Benchmark Models						Regularized Models			Tree-Based Models			Neural Networks							
	HAR	**HAR-X**	**LogHAR**	**LevHAR**	**SHAR**	**HARQ**	**RR**	**LA**	**EN**	**BG**	**RF**	**GB**	$\mathbf{NN}_1^1$	$\mathbf{NN}_1^{10}$	$\mathbf{NN}_2^1$	$\mathbf{NN}_2^{10}$	$\mathbf{NN}_3^1$	$\mathbf{NN}_3^{10}$	$\mathbf{NN}_4^1$	$\mathbf{NN}_4^{10}$
HAR	–	1.000	0.936**	0.951	1.031	0.846**	1.028	1.039	1.039	0.941	0.916*	0.973	1.053	1.017	1.038	1.033	1.063	1.052	1.058	1.061
HAR-X	1.000	–	0.936**	0.951	1.031	0.846**	1.028	1.039	1.039	0.941	0.916*	0.973	1.053	1.017	1.038	1.033	1.063	1.052	1.058	1.061
LogHAR	1.068	1.068	–	1.016	1.101	0.904**	1.098	1.110	1.110	1.005	0.979	1.040	1.125	1.087	1.109	1.104	1.136	1.124	1.130	1.134
LevHAR	1.052	1.052	0.984	–	1.084	0.890**	1.081	1.093	1.093	0.989	0.964	1.024	1.108	1.070	1.092	1.086	1.118	1.106	1.113	1.116
SHAR	0.970	0.970	0.908*	0.922	–	0.821**	0.997	1.008	1.008	0.913	0.889*	0.944	1.022	0.987	1.007	1.002	1.031	1.020	1.026	1.029
HARQ	1.182	1.182	1.106	1.124	1.218	–	1.215	1.228	1.228	1.112	1.083	1.150	1.245	1.202	1.227	1.221	1.256	1.243	1.250	1.254
RR	0.973**	0.973**	0.911***	0.925*	1.003	0.823***	–	1.011	1.011	0.915	0.892**	0.947	1.025	0.990	1.010	1.005	1.034	1.023	1.030	1.032
LA	0.962***	0.962***	0.901***	0.915**	0.992	0.814***	0.989***	–	1.000*	0.905*	0.882**	0.937*	1.013	0.979	0.999	0.994	1.023	1.012	1.018	1.021
EN	0.963***	0.963***	0.901***	0.915**	0.992	0.815***	0.989***	1.000	–	0.905*	0.882**	0.937*	1.014	0.979	0.999	0.994	1.023	1.012	1.018	1.021
BG	1.063	1.063	0.995	1.011	1.096	0.900*	1.093	1.105	1.104	–	0.974	1.035	1.120	1.082	1.104	1.098	1.130	1.118	1.125	1.128
RF	1.091	1.091	1.022	1.038	1.125	0.924*	1.122	1.134	1.134	1.027	–	1.062	1.149	1.110	1.133	1.127	1.160	1.148	1.155	1.158
GB	1.027	1.027	0.962*	0.977	1.059	0.869***	1.056	1.068	1.067	0.966	0.941**	–	1.082	1.045	1.067	1.061	1.092	1.081	1.087	1.090
$\mathbf{NN}_1^1$	0.950	0.950	0.889***	0.903	0.979	0.803**	0.976	0.987	0.986	0.893	0.870**	0.924*	–	0.966	0.986*	0.981**	1.009	0.999	1.005	1.007
$\mathbf{NN}_1^{10}$	0.983	0.983	0.920***	0.935	1.013	0.832**	1.010	1.022	1.021	0.925	0.901**	0.957	1.035	–	1.020	1.015	1.045	1.034	1.040	1.043
$\mathbf{NN}_2^1$	0.963	0.963	0.902**	0.916	0.993	0.815**	0.990	1.001	1.001	0.906	0.883*	0.938	1.015	0.980	–	0.995	1.024	1.013	1.019	1.022
$\mathbf{NN}_2^{10}$	0.968	0.968	0.906**	0.921	0.998	0.819**	0.995	1.006	1.006	0.911	0.887*	0.942	1.020	0.985	1.005	–	1.029	1.018	1.024	1.027
$\mathbf{NN}_3^1$	0.941	0.941	0.881**	0.895	0.970	0.796**	0.967	0.978	0.977	0.885	0.862**	0.916*	0.991	0.957*	0.977***	0.972**	–	0.990**	0.995	0.998
$\mathbf{NN}_3^{10}$	0.951	0.951	0.890**	0.904	0.980	0.805**	0.977	0.988	0.988	0.894	0.871**	0.925	1.001	0.967	0.987***	0.982*	1.011	–	1.006	1.009
$\mathbf{NN}_4^1$	0.945	0.945	0.885**	0.899	0.974	0.800**	0.971	0.982	0.982	0.889	0.866*	0.920	0.995	0.962	0.981	0.976	1.005	0.994	–	1.003
$\mathbf{NN}_4^{10}$	0.943	0.943	0.882**	0.896	0.972	0.798**	0.969	0.980	0.979	0.887	0.864**	0.917	0.993	0.959	0.978**	0.974*	1.002	0.991	0.997	–

Notes: Each entry (i, j) reports the out-of-sample MSE of row model i relative to column model j, computed over the test period 2016-04-21 to 2020-08-06 (1,099 observations). A value below unity indicates that column model j outperforms row model i. Significance stars denote rejection of equal predictive accuracy in favour of the column model based on one-sided Diebold and Mariano (2002) tests with Newey–West standard errors (5 lags). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 6: Cleveland Dot Plot of $\mathcal{M}_{\text{HAR}}$ Results



Notes: The Cleveland dot plot reports each model's out-of-sample relative MSE relative to the standard HAR model, which is normalized to unity, under the $\mathcal{M}_{\text{HAR}}$ feature set. Values below 1 indicate that the model outperforms the HAR model, while values above 1 indicate the opposite. Models are ordered from best to worst according to their relative MSE with respect to the HAR model. Blue markers denote models with lower relative MSE than HAR; grey markers denote the benchmark-equivalent HAR and HAR-X specifications, and red markers denote models with higher relative MSE than HAR. The numerical labels next to each marker report the exact relative MSE, while the significance stars indicate the outcome of the one-sided Diebold and Mariano (2002) test against HAR: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

5.2 Results from $\mathcal{M}_{\text{ALL}}$ Feature Set

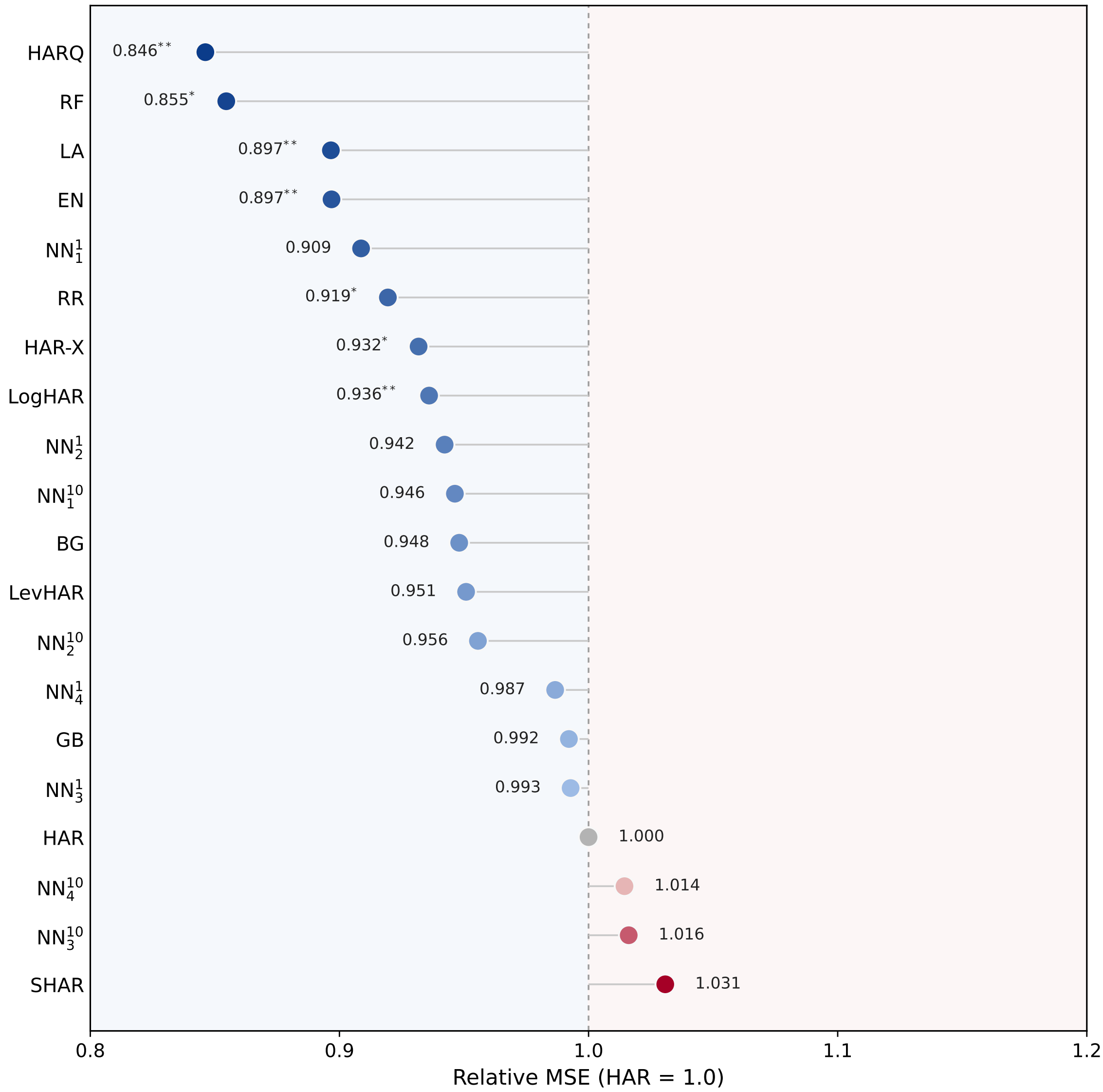
In this segment, we report the empirical results for the $\mathcal{M}_{\text{ALL}}$ feature set. Table (4) illustrates the pairwise relative MSE performance of each model pair. Moreover, we conduct the Diebold and Mariano (2002) test to observe whether the difference in model performances is statistically significant or not. In Figure (7) we illustrate the relative MSE relative to the standard HAR model, with ranked performance from best to worst model for ease of interpretation.

Table 4: Pairwise Relative MSE and Diebold-Mariano Tests on $\mathcal{M}_{\mathrm{ALL}}$ Feature Set

	Benchmark Models						**Regularized Models**			**Tree-Based Models**			**Neural Networks**							
	HAR	**HAR-X**	**LogHAR**	**LevHAR**	**SHAR**	**HARQ**	**RR**	**LA**	**EN**	**BG**	**RF**	**GB**	$\mathbf{NN}_1^1$	$\mathbf{NN}_1^{10}$	$\mathbf{NN}_2^1$	$\mathbf{NN}_2^{10}$	$\mathbf{NN}_3^1$	$\mathbf{NN}_3^{10}$	$\mathbf{NN}_4^1$	$\mathbf{NN}_4^{10}$
HAR	–	0.932*	0.936**	0.951	1.031	0.846**	0.919*	0.897**	0.897**	0.948	0.855*	0.992	0.909	0.946	0.942	0.956	0.993	1.016	0.987	1.014
HAR-X	1.073	–	1.005	1.020	1.106	0.908**	0.987	0.962*	0.962*	1.017	0.917	1.065	0.975	1.016	1.011	1.026	1.065	1.091	1.059	1.089
LogHAR	1.068	0.996	–	1.016	1.101	0.904**	0.982	0.958**	0.958**	1.013	0.913	1.060	0.971	1.011	1.007	1.021	1.061	1.086	1.054	1.084
LevHAR	1.052	0.980	0.984	–	1.084	0.890**	0.967	0.943*	0.943*	0.997	0.899*	1.043	0.956	0.995	0.991	1.005	1.044	1.069	1.038	1.067
SHAR	0.970	0.904*	0.908*	0.922	–	0.821**	0.892*	0.870**	0.870**	0.920	0.829*	0.962	0.882	0.918	0.914	0.927	0.963	0.986	0.957	0.984
HARQ	1.182	1.101	1.106	1.124	1.218	–	1.087	1.060	1.060	1.120	1.010	1.172	1.074	1.118	1.113	1.129	1.173	1.201	1.166	1.199
RR	1.088	1.013	1.018	1.034	1.121	0.920*	–	0.975*	0.975*	1.031	0.929	1.079	0.988	1.029	1.025	1.039	1.080	1.105	1.073	1.103
LA	1.115	1.039	1.044	1.061	1.150	0.944	1.026	–	1.000	1.057	0.953	1.107	1.014	1.056	1.051	1.066	1.107	1.133	1.100	1.131
EN	1.115	1.039	1.044	1.060	1.149	0.944	1.025	1.000**	–	1.057	0.953	1.106	1.013	1.055	1.051	1.066	1.107	1.133	1.100	1.131
BG	1.055	0.983	0.987	1.003	1.087	0.893	0.970	0.946	0.946	–	0.901**	1.046	0.958	0.998	0.994	1.008	1.047	1.072	1.041	1.070
RF	1.170	1.090	1.095	1.113	1.206	0.990	1.076	1.049	1.049	1.109	–	1.161	1.063	1.107	1.103	1.118	1.162	1.189	1.154	1.187
GB	1.008	0.939	0.943	0.958	1.039	0.853	0.927	0.904	0.904	0.956	0.861	–	0.916	0.954	0.950	0.963	1.001	1.024	0.994	1.022
$\mathbf{NN}_1^1$	1.100	1.025	1.030	1.046	1.134	0.931**	1.012	0.987	0.987	1.043	0.940	1.092	–	1.041	1.037	1.052	1.093	1.118	1.086	1.116
$\mathbf{NN}_1^{10}$	1.057	0.985	0.989	1.005	1.089	0.894*	0.972	0.947*	0.948*	1.002	0.903	1.048	0.960	–	0.996	1.010	1.049	1.074	1.043	1.072
$\mathbf{NN}_2^1$	1.061	0.989	0.993	1.009	1.094	0.898**	0.976	0.952**	0.952**	1.006	0.907	1.053	0.964	1.004	–	1.014	1.054	1.078	1.047	1.077
$\mathbf{NN}_2^{10}$	1.046	0.975	0.979	0.995	1.079	0.886*	0.962	0.938**	0.939**	0.992	0.894	1.038	0.951	0.990	0.986	–	1.039	1.063	1.032	1.062
$\mathbf{NN}_3^1$	1.007	0.939*	0.943**	0.958	1.038	0.852**	0.926**	0.903***	0.903***	0.955	0.861*	0.999	0.915*	0.953***	0.949***	0.963***	–	1.023	0.994	1.022
$\mathbf{NN}_3^{10}$	0.984	0.917*	0.921**	0.936	1.014	0.833**	0.905**	0.882***	0.883***	0.933	0.841*	0.976	0.894*	0.931***	0.927***	0.940***	0.977	–	0.971***	0.998
$\mathbf{NN}_4^1$	1.014	0.944	0.949*	0.964	1.045	0.858**	0.932**	0.909**	0.909**	0.961	0.866	1.006	0.921	0.959**	0.955**	0.969***	1.006	1.030	–	1.028
$\mathbf{NN}_4^{10}$	0.986	0.919*	0.923**	0.937	1.016	0.834**	0.906**	0.884***	0.884***	0.935	0.842*	0.978	0.896*	0.933***	0.929***	0.942***	0.979	1.002	0.973**	–

Notes: Each entry (i, j) reports the out-of-sample MSE of row model i relative to column model j, computed over the test period 2016-04-21 to 2020-08-06 (1,099 observations). A value below unity indicates that column model j outperforms row model i. Significance stars denote rejection of equal predictive accuracy in favour of the column model based on one-sided Diebold and Mariano (2002) tests with Newey–West standard errors (5 lags). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 7: Cleveland Dot Plot of $\mathcal{M}_{\text{ALL}}$ Results



Notes: The Cleveland dot plot reports each model's out-of-sample relative MSE relative to the standard HAR model, which is normalized to unity, under the $\mathcal{M}_{\text{ALL}}$ feature set. Values below 1 indicate that the model outperforms the HAR model, while values above 1 indicate the opposite. Models are ordered from best to worst according to their relative MSE with respect to the HAR model. Blue markers denote models with lower relative MSE than HAR; grey markers denote the benchmark-equivalent HAR specification, and red markers denote models with higher relative MSE than HAR. The numerical labels next to each marker report the exact relative MSE, while the significance stars indicate the outcome of the one-sided Diebold and Mariano (2002) test against HAR: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

6 Discussion

6.1 Limitations

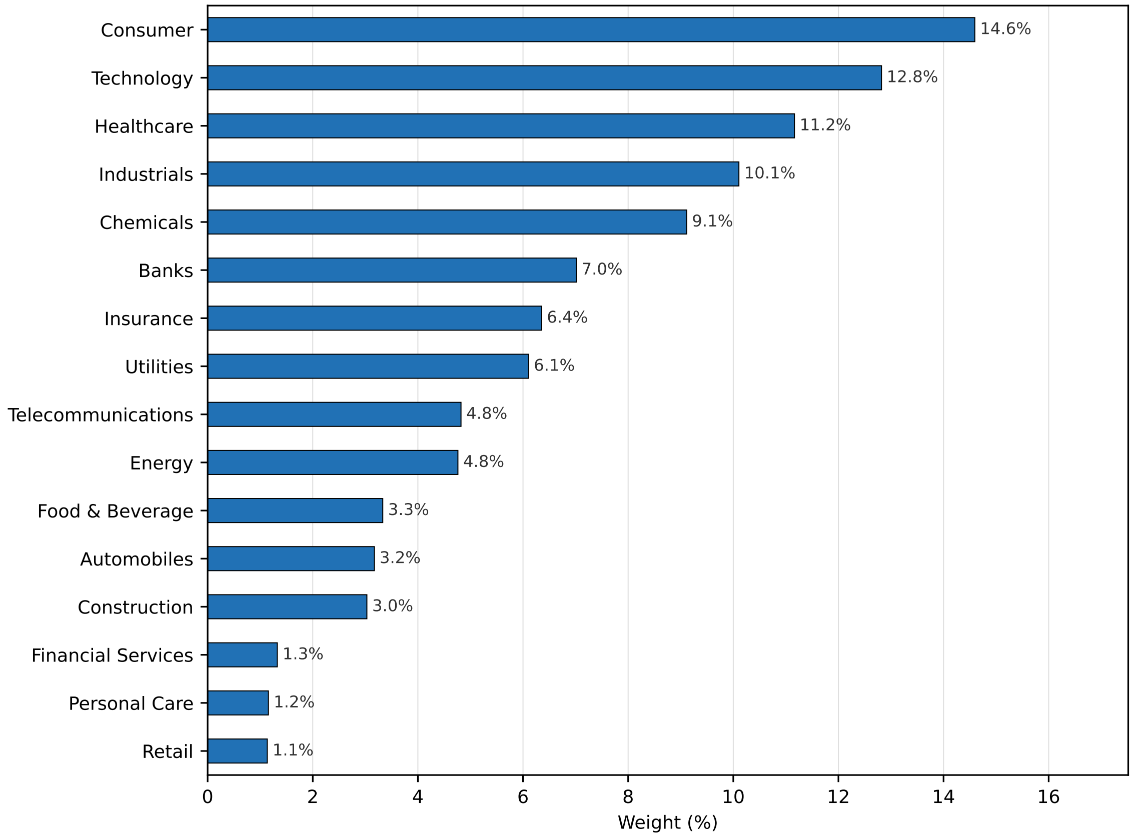
6.2 Future Research

7 Concluding Remarks

8 Appendix

8.1 EURO STOXX 50 Industry Composition

Figure 8: EURO STOXX 50 Industry Sector Composition (2020)



Notes: The figure plots the industry sector composition for the EURO STOXX 50 index as of June 30, 2020. Please note that this is only a snapshot in time, because sector industry composition data is only available on a quarterly basis from 2019 onwards. As such, the weights are not representative for our entire sample period, yet provide a general overview of the EURO STOXX 50 index with weights expected to remain fairly stable across our sample. Index constituents, their respective sector classification, and weights are sourced from the STOXX website.

8.2 Volatile Events

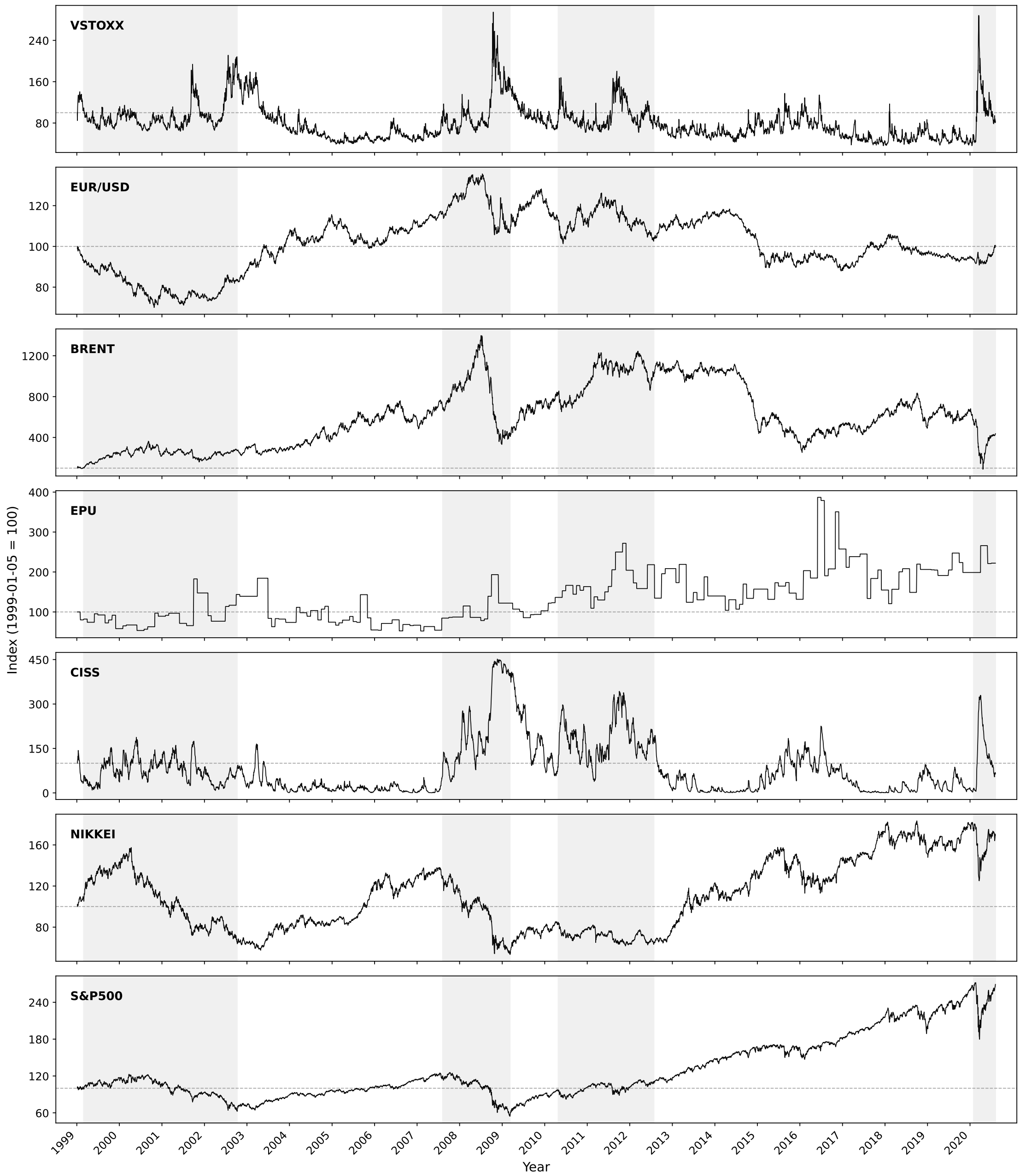
Table 5: Description of Annual Volatile Event Outliers on the EURO STOXX 50

Year	**Date**	**Description of Volatile Event**
1999	Jan. 15	Brazil floated the Real, abandoning its currency peg.
2000	Apr. 17	Dot-com bubble burst; NASDAQ recorded one of its steepest weekly declines.
2001	Sep. 11	Al-Qaeda attacks on the World Trade Center and the Pentagon.
2002	Jul. 25	WorldCom's bankruptcy filing amid widespread U.S. accounting scandals.
2003	Mar. 17	U.S-led coalition finalised preparations for the invasion of Iraq.
2004	Mar. 11	Al-Qaeda bombings on Madrid's commuter rail network.
2005	Jul. 07	Four coordinated suicide bombings on London's transport network.
2006	May 18	Rising U.S. inflation fears and Fed rate hike expectations.
2007	Aug. 17	The Fed made an emergency discount rate cut as credit markets froze.
2008	Oct. 10	Post-Lehman crisis peak; equity markets worst weekly performance since 1929.
2009	Jan. 23	Continued banking sector deterioration and deepening recession fears.
2010	May 07	Greece's emergency IMF-EU bailout intensified sovereign debt contagion fears.
2011	Aug. 11	S&P's U.S. downgrade and escalating Italian and Spanish debt contagion.
2012	Aug. 02	Draghi's *"Whatever it takes"* pledge (Jul. 26)
2013	Feb. 25	Inconclusive Italian elections yielded a hung parliament.
2014	Oct. 16	Downward IMF growth revisions, falling oil prices, and Ebola fears.
2015	Aug. 24	China's Shanghai Composite fell 8%+ ("Black Monday").
2016	Jun. 24	Announcement of the Brexit referendum results.
2017	Apr. 24	The French presidential election first round generated fears.
2018	Feb. 06	A VIX spike *"Volmageddon"* collapsed short-volatility products.
2019	Aug. 15	U.S-China trade war escalation and yuan depreciation past 7 per dollar.
2020	Mar. 16	COVID-19 pandemic lockdowns across Europe.

Notes: This table provides an event descirption for each of the volatile event outliers presented in the boxplot in Figure (3) for the entire sample period from the 5th of January to the 6th of August 2020.

8.3 Indexed Evolution of the Macroeconomic Predictors

Figure 9: Macroeconomic Variables Panel Included in $\mathcal{M}_{\mathrm{ALL}}$ Feature Set



Notes: This figure presents the macroeconomic variables included in the $\mathcal{M}_{\mathrm{ALL}}$ feature set. The index for each variable is set at 100 at the start of the sample period, on the 5th of January 1999. The sample period runs from the 5th of January 1999 to the 6th of August 2020. Shaded regions indicate significant and persistent volatility regimes, previously described in Figure (**??**). These are, from left to right, the Dot-com Bubble (2000 - 2002), the Global Financial Crisis (GFC, 2008 - 2009), the European Sovereign Debt Crisis (Euro Crisis, 2010 - 2012), and the Covid-19 pandemic (2020).

8.4 EURO STOXX 50 Index Constituents

Table 6: An Overview of the EURO STOXX 50 Index Constituents (2020)

Company	Sector	Country	Weight (%)
ASML HLDG	Technology	Netherlands	6.025
SAP	Technology	Germany	5.890
LINDE	Chemicals	Germany	4.498
LVMH MOET HENNESSY	Consumer	France	4.492
SANOFI	Healthcare	France	4.466
TOTAL	Energy	France	3.828
SIEMENS	Industrials	Germany	3.394
ALLIANZ	Insurance	Germany	3.283
L'OREAL	Consumer	France	3.014
UNILEVER NV	Consumer	Netherlands	2.806
BAYER	Healthcare	Germany	2.799
IBERDROLA	Utilities	Spain	2.637
AIR LIQUIDE	Chemicals	France	2.632
ENEL	Utilities	Italy	2.583
SCHNEIDER ELECTRIC	Industrials	France	2.359
DEUTSCHE TELEKOM	Telecommunications	Germany	2.098
BASF	Chemicals	Germany	1.984
VINCI	Construction	France	1.974
ADIDAS	Consumer	Germany	1.863
BNP PARIBAS	Banks	France	1.766
DANONE	Food & Beverage	France	1.730
AXA	Insurance	France	1.667
PHILIPS	Healthcare	Netherlands	1.646
ANHEUSER-BUSCH INBEV	Food & Beverage	Belgium	1.603
AIRBUS	Industrials	France	1.595
BCO SANTANDER	Banks	Spain	1.565
KERING	Consumer	France	1.563
ESSILORLUXOTTICA	Healthcare	France	1.467
MUENCHENER RUECK	Insurance	Germany	1.404
DEUTSCHE POST	Industrials	Germany	1.385
SAFRAN	Industrials	France	1.374
DEUTSCHE BOERSE	Financial Services	Germany	1.325
DAIMLER	Automobiles	Germany	1.314
INTESA SANPAOLO	Banks	Italy	1.204
AHOLD DELHAIZE	Personal Care	Netherlands	1.156
INDUSTRIA DE DISENO TEXTIL SA	Retail	Spain	1.134
VOLKSWAGEN PREF	Automobiles	Germany	1.070
CRH	Construction	Ireland	1.056
ING GRP	Banks	Netherlands	1.047
NOKIA	Telecommunications	Finland	0.952
ENI	Energy	Italy	0.934
ORANGE	Telecommunications	France	0.913
AMADEUS IT GROUP	Technology	Spain	0.905
ENGIE	Utilities	France	0.886
BCO BILBAO VIZCAYA ARGENTARIA	Banks	Spain	0.885
TELEFONICA	Telecommunications	Spain	0.857
VIVENDI	Consumer	France	0.855
BMW	Automobiles	Germany	0.788
FRESENIUS	Healthcare	Germany	0.785
GRP SOCIETE GENERALE	Banks	France	0.547

Notes: This table reports the constituents of the EURO STOXX 50 index. The constituent weights are as of June 30, 2020 and sourced from STOXX Ltd. It should be noted that the composition of the index is subject to additions and deletions, which is not accounted for in this table. The table provides an illustrative overview of the types of companies included at the end of our sample period.

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